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Master Thesis

Analyzing amphibian metacommunities in river-floodplain systems using eDNA metabarcoding:

The importance of spatial connectivity and environmental conditions

submitted by

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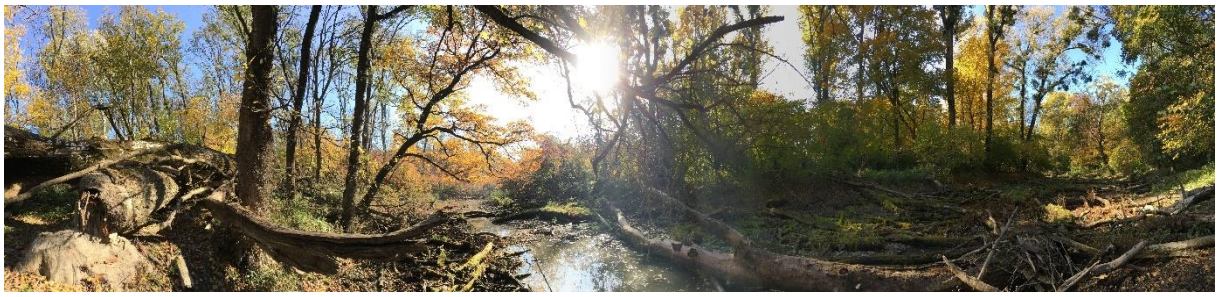
Master Thesis



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Affidavit

I hereby declare that I have authored this master thesis independently, and that I have not used any assistance other than that which is permitted. The work contained herein is my own except where explicitly stated otherwise. All ideas taken in wording or in basic content from unpublished sources or from published literature, as well as those which were generated using artificial intelligence tools, are duly identified and cited, and the precise references included.

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Vienna, 20.01.2025 Peter BADER *(manu propria)*

This thesis is dedicated to my grandma Hedwig and grandpa Karlheinz Bader
and my parents Kathrine (Tina) and Karlheinz Bader

The mystery of life is not a problem to be solved;
it is a reality to be experienced

Jacobus Johannes Leeuw, The Conquest of Illusion, 1928

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Abstract

Riverine ecosystems are hotspots for biodiversity but are threatened by anthropogenic alteration and climate change. The Danube National Park floodplain in Austria hosts 67% of the 21 native amphibian species in the region (14 anurans, 7 caudates), making it a valuable area to study metacommunities for conservation and restoration efforts. As part of the RIMECO project, this study assesses the diversity and composition of an amphibian metacommunity to evaluate the relative importance of local environmental conditions and spatial connectivity in its assembly. In summer 2021, environmental DNA (eDNA) samples were collected twice by water filtration from 26 water bodies while determining environmental conditions in the National Park Donau-Auen, located between Vienna and Bratislava (Austria), one of the last free-flowing sections of the Danube River. The collected data was used for metacommunity analyses, marking this the first use of eDNA metabarcoding in altered floodplain ecosystems. By using this innovative method for community detection and explicitly including spatial connectivity, a deeper understanding of managing protected communities in fragmented river ecosystems is expected. The results showed detections of seven amphibian taxa (5 anurans, 2 caudates), with *Lissotriton vulgaris*, *Pelobates fuscus*, and *Bombina bombina* being correlated with shallow, small water bodies with high chlorophyll-a concentrations. *Pelophylax* was the most widespread species and negatively correlated by fish presence, whereas *Bufo bufo* appeared to be associated with fish. Overall, amphibian metacommunities seem to be driven by the interaction of local and regional processes, with the most diverse communities found in isolated water bodies with low connectivity to the river main channel. Variation in their composition is best explained by combining environmental, connectivity, and biotic variables, emphasizing their importance in managing and restoring fragmented river ecosystems.

Keywords: environmental DNA (eDNA) metabarcoding, riverine floodplain ecosystems, amphibians, metacommunity, spatial connectivity, environmental conditions, Danube River

Kurzfassung

Flusssysteme sind Biodiversitäts-Hotspots aber durch anthropogene Einflüsse und den Klimawandel bedroht. Die Flussauen im Donau-Auen Nationalpark in Österreich beherbergt 67% der 21 heimischen Amphibien (14 Frosch- und 7 Schwanzlurche) und stellt ein wertvolles Gebiet für die Untersuchung von Metagemeinschaften (MG) im Rahmen von Schutz- und Restaurierungsmaßnahmen dar. Als Teil des RIMECO-Projekts bewertet diese Studie die Diversität und Zusammensetzung einer Amphibien-Metagemeinschaft (AMG), um die Bedeutung von lokalen Umweltbedingungen und räumlicher Konnektivität zu untersuchen. Im Sommer 2021 wurden zweimal eDNA-Proben (Umwelt-DNA) durch Wasserfiltration aus 26 Gewässern entnommen und Umweltbedingungen im Nationalpark Donau-Auen zwischen Wien und Bratislava (AUT) erfasst, einem der letzten frei fließenden Abschnitte der Donau. Die gewonnenen Daten wurden für MG-Analysen verwendet, was den ersten Einsatz von eDNA-Metabarcoding in veränderten Flussauen darstellt. Diese innovative Methode zur Erfassung von Biozönosen und explizite Einbeziehung räumlicher Konnektivität soll ein besseres Verständnis für das Management geschützter Lebensgemeinschaften in fragmentierten Flussökosystemen fördern. Die Ergebnisse zeigen Nachweise von sieben Amphibien-Taxa (5 Frosch- und 2 Schwanzlurche), wobei *Lissotriton vulgaris*, *Pelobates fuscus* und *Bombina bombina* mit flachen, kleinen Gewässern und hoher Chlorophyll-a-Konzentration korrelierten. *Pelophylax* war am weitesten verbreitet und negativ mit Fischen korreliert, während *Bufo bufo* mit Fischen assoziiert war. Insgesamt scheinen AMG durch die Interaktion lokaler und regionaler Prozesse geprägt, wobei die größte Diversität in isolierten Gewässern mit geringer Anbindung zum Fluss gefunden wurde. Die Variation in ihrer Zusammensetzung lässt sich am besten durch die Kombination von Umwelt-, Konnektivitäts- und biotischen Variablen erklären, was deren Bedeutung für das Management fragmentierter Flussökosysteme unterstreicht.

Keywords: Umwelt DNA (eDNA) Metabarcoding, Fluss-Auen Ökosysteme, Amphibien, Metagemeinschaft, Räumliche Konnektivität, Umweltbedingungen, Donau

Introduction

Riverine floodplain systems are extraordinarily complex and highly diverse biospheres (Funk et al., 2021; Shiel et al., 1998) which provide a vast number of essential ecosystem functions and services (Funk et al., 2021; Hein et al., 2006; Sanon et al., 2012). Due to land use changes and engineering measures like habitat alteration or flow and flood control, they are among the most endangered ecosystems (Dudgeon et al., 2006; Tockner & Stanford, 2002). This is expressed in the form of loss of floodplain areas, riparian biotopes, and alteration of hydro-morphological conditions, particularly in Europe (Habersack et al., 2016; Preiner et al., 2018; Revenga, 2000; Thomsen et al., 2012). Thus, there is an urgent need to conserve, restore, and manage these vulnerable systems. To achieve this, we rely on comprehensive monitoring methods to understand their dynamics and functions (Cardinale et al., 2012; Funk et al., 2019; Schindler et al., 2016; Tickner et al., 2020).

Studying amphibians contributes substantially to conceptualizing the functioning of dynamic floodplain ecosystems due to their semiaquatic lifestyle. This is also what makes these animals particularly vulnerable, and thus, the most threatened group according to the IUCN Red List of Threatened Species (IUCN Red List of Threatened Species, 2024; Nori et al., 2015). Amphibians use dynamic landscapes and are dependent on aquatic areas they disperse from (Kačergytė et al., 2021; Tockner et al., 2006). Hence, they are an important subject for metacommunity studies (Jehle, 1997). Wilson (1992) describes a metacommunity as a network of potentially interacting local communities, which are linked by dispersal. This is a key concept in community ecology and mainly addresses patterns of species distribution, abundance, and interactions on the level of local to regional scales (Chase et al., 2020; Leibold et al., 2004; Schlägel et al., 2020) as depicted in Figure 1. It includes patch dynamics, species sorting (niche concept), mass effect (source-sink dynamics) and the neutral model as the four classical fundamental theoretical frameworks (Hubbell, 2004; Leibold et al., 2004, 2021; Schlägel et al., 2020). Leibold et al. (2022) developed an alternative, more direct conceptual framework to understand the impact of species and patch heterogeneities on the structure of metacommunities (see Figure 2). They provided a summary on how basic mechanisms (e.g., species interactions, food web structure, biogeography, phylogeny, microevolution) drive processes of community assembly (e.g., drift, selection, dispersal, speciation) that directly shape biodiversity patterns. Studying these created patterns can help explain the overall metacommunity structure.

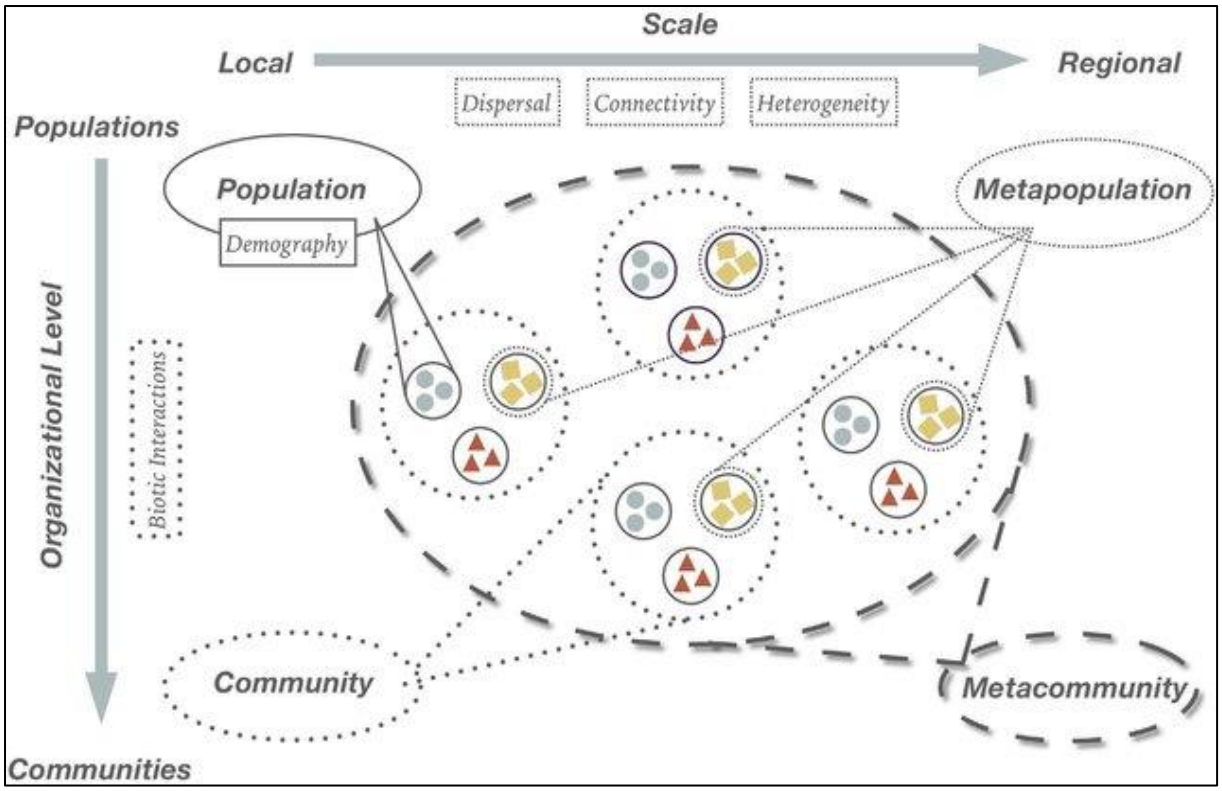

Figure 1: Concept of metacommunities. Figure reprinted from Chase et al. (2020).

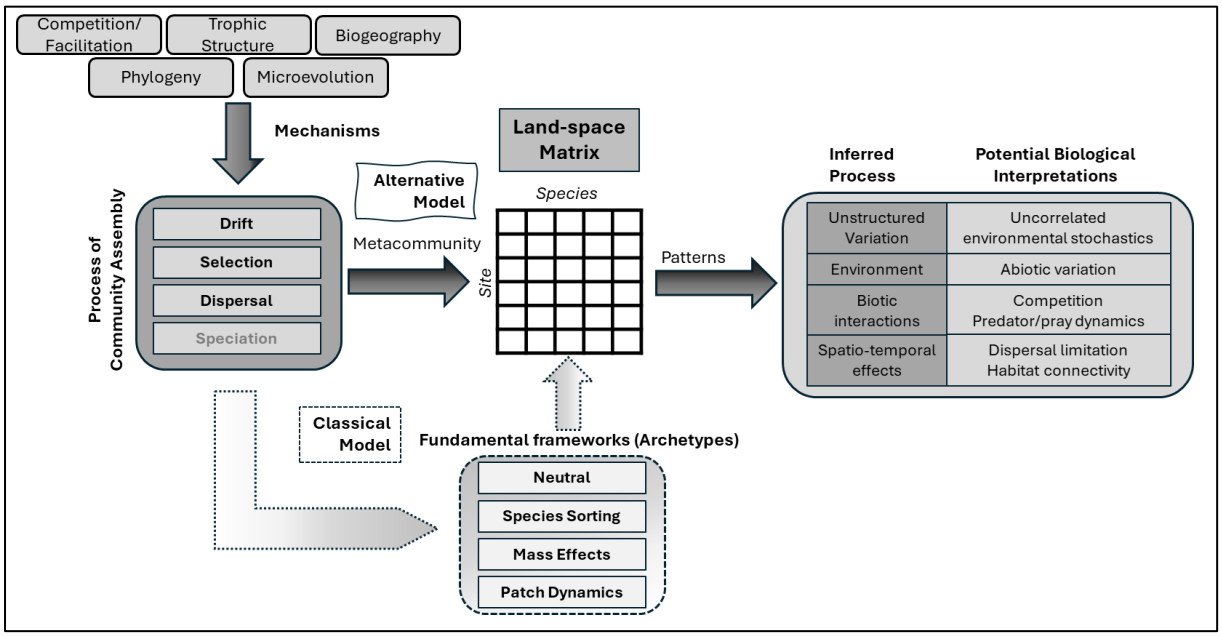

Figure 2: Summary of concepts in metacommunity ecology after Leibold et al. (2022). The upper section shows the alternative more direct model: Basic mechanisms (e.g., species interactions, food web structure, biogeography, phylogeny, microevolution) influence processes of community assembly (e.g., drift, selection, dispersal) which shape the metacommunity structure in a land-space matrix. Observable patterns of these biodiversity structures can be used to explain the underlying effects of the overall metacommunity structure. The lower section displays the classical more indirect and idealized model from Leibold et al. (2004). Graphic adapted from Leibold et al. (2022).

Amoros & Bornette (2002) describe that riverscapes like floodplains are characterized by high α-, β-, and γ-diversity due to a combination of complex gradients of habitat conditions and hydrological connectivity. Furthermore, they state that α-diversity patterns differ between biota within a floodplain metacommunity (see Figure 3 and Tockner et al., 1998). For example, they show that α-diversity of fish increases with hydrological connectivity, while amphibians show the exact opposite pattern. Alford & Richards (1999), as well as Marsh & Trenham (2001), stated that most amphibians show a metapopulation structure, which Tockner et al. (2006) found to differ between active floodplains and riparian forests.

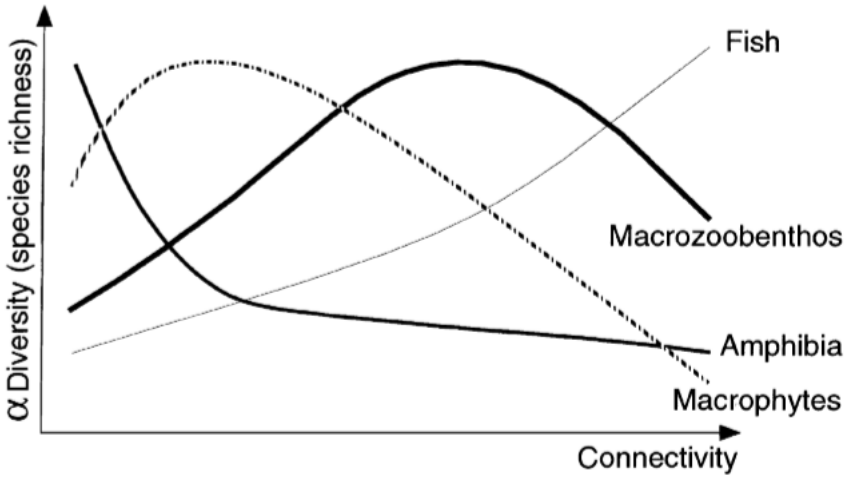

Figure 3: Idealized patterns of alpha-diversity (species richness) of different taxonomic groups according to river connectivity in an Austrian floodplain of the Danube River. Reprinted from Amoros & Bornette (2002).

On a local scale, environmental conditions determine habitat suitability (the fundamental niche) and therefore the occurrence of a species (realized niche) (Davis et al., 2016). These characteristics can further be divided into (1) biotic parameters like interspecific (e.g., predator-prey interaction), as well as intraspecific competition (Davis et al., 2016), and (2) abiotic characteristics of the environment, which both define the fundamental niche (niche dimensions; see Schaefer, 2012, pp. 193ff.). Of these factors, hydroperiod, patch size (surface area), aquatic vegetation, canopy-cover, local population size or biomass, and predation seem to be the most important to explain amphibian metacommunity structure and dynamics on a local scale (Davis et al., 2017; Gabrielsen et al., 2022; Hamer et al., 2021; Kačergytė et al., 2021; Nagel et al., 2021; Semlitsch, 2000; Vági et al., 2013; Werner et al., 2007a,b).

For example, aquatic vegetation provides hiding and spawning areas for amphibians (Kačergytė et al., 2021), while canopy cover provides important terrestrial habitat (Hamer et al., 2021) and causes shading which, hence, can influence air and water temperature. Werner et al. (2007a,b), as well as Semlitsch (2000), stress the importance of population size (biomass) for species turnover and source-sink processes. However, presence of, and therefore predation by fish is found to be one of the most important biological factors influencing amphibian metacommunity structure (Hamer et al., 2021; Johnson et al., 2013). Laufer & Wollenzin (2011) gave an overview of possible effects of fish presence on amphibians (see Table 1). For example (see Table 6)*, Perca fluviatilis* (European perch) feed on most amphibians like *Bufo bufo* (L. 1758), *Pelophylax* kl. *esculentus* (L. 1758), *Rana temporaria* (L. 1758), and even on *Pelobates fuscus* (LAURENTI 1768). *Rutilus rutilus* (roach) also often prey on the larvae of *P.* kl. *esculentus*, and *R. temporaria* but rarely of *B. bufo*. *Alburnus alburnus* (common bleak) on the other hand feed on *R. temporaria* but not on the genus *Pelophylax* or *B. bufo. Scardinius erythrophthalmus* (rudd) only rarely prey on larvae of *B. bufo* but often those of *Lissotriton vulgaris* (LINNAEUS 1758), *P. esculentus,* and *R. temporaria*. Also, species of the genus *Gymnocephalus* (i.e., *G. cernua*) are thought to feed on amphibian larvae (Gebhardt, 1983).

Table 1: Overview-table of selected amphibians and their risk from fish, adapted from Laufer & Wollenzin (2011).

Species [1)]	Risk from fish [2)]	Literature (selection)
Great crested newt ***Triturus cristatus***	+++	Gebhardt, 1983; Ortmann, 2009; Bardal et al., 2010; Wright, 2010; Kuhn, 2001
Smooth newt *Lissotriton vulgaris*	++	Breuer 1992, Gebhardt 1983, Hehmann & Zucchi 1985, Meier 1995, Denoël et al. 2009, Ortmann 2009, Bardal et al. 2010, Drobny 2006
Fire-bellied toad ***Bombina bombina***	+++	
Common speadfoot ***Pelobates fuscus***	++	Flindt & Hemmer, 1969; Kuzmin, 1999
Common toad *Bufo bufo*	+	Breuer, 1992; Gebhardt, 1983; Bauser et al., 1987; Glandt, 1983; Glandt, 1984; Hehmann & Zucchi, 1985; Spolwind et al., 2001; Clausnitzer, 1983; Laufer, 2007
Green toad ***Bufotes viridis***	+++	Breuer, 1992; Gebhardt, 1983; Hehmann & Zucchi, 1985; Niekisch, 1995; Flindt & Hemmer, 1969; Winkler & Dierking, 2003
European tree frog ***Hyla arborea***	+++	Meier, 1995; Pastors, 1995; Clausnitzer & Berninghausen, 1991; Brandt, 2007; Filoda, 1981; Glandt, 2004a,b; Brönmark & Edenhamn, 1994; Tester, 1990; Clausnitzer, 2010; Kuhn, 1998; Drobny, 2006
Moor frog ***Rana arvalis***	++	Filoda, 1981; Gebhardt, 1983
Agile frog ***Rana dalmatina***	++	Spolwind & Pintar, 1997; Laufer et al., 1997
Common or grass frog *Rana temporaria*	++	Breuer, 1992; Leu et al., 2009; Gebhardt, 1983; Heusser & Schlumpf, 1971; Meisterhans & Heusser, 1970; Broggi, 1975; Kabisch & Weiss, 1968; Kleisinger & Wagner, 1993; Glandt, 1983; Glandt, 1984; Glandt, 1985; Escher, 1972; Honegger, 1978; Hehmann & Zucchi, 1985; Flindt & Hemmer, 1969; Laufer, 2007; Schay, 1988; Loman, 2002
Water frogs *Pelophylax* sp.	++	Semlitsch, 1993; Gabhardt, 1983; Flindt & Hemmer, 1969
Edible frog *Pelophylax* kl. *esculentus*	++	Breuer, 1992; Gebhardt, 1983; Glandt, 1983, Glandt, 1984; Semlitsch, 1993; Hehman & Zucchi, 1985
Pool frog ***Pelophylax lessonae***	++	
Lake frog *Pelophylax ridibundus*	+	

+ low risk from fishes
++ medium risk from fishes
+++ high risk from fishes, which can lead to the extinction of the stock in the water body
1) Species marked in bold are strictly protected under European law
2) Professional assessment based on the literature or plausibility

On a regional scale, spatial connectivity is the main determinant of habitat occupancy (hence the realized niche) and whether dispersal (or accessibility via mobility) is favored or limited (Chase et al., 2020). The latter is dependent on several factors such as urbanization (e.g., road density, land cover proportion and types, as well as their distances to habitats), surrounding terrestrial matrix (e.g., density of vegetation or forest cover and their distance to waters), climate (e.g., annual precipitation patterns), linkage of waterbodies (distance and connectivity), and total area of the wetlands (Davis et al., 2017; Hamer et al., 2021; Johnson et al., 2013; Richter-Boix et al., 2007; Werner et al., 2007a,b). As a surrogate parameter, hydrological connectivity (i.e., linkage to the main channel in days per year or in percentages) can be used, which describes the gradient from lentic to lotic conditions and how often these conditions change in floodplain water bodies (Hohensinner et al., 2011; Tockner & Ward, 1999). In more detail, water bodies can be grouped topologically in Eupotamon, Parapotamon A or B, Plesiopotamon, and Paleopotamon (Hohensinner et al., 2011; Tockner & Ward, 1999), hereinafter referred to as highly connected (Eupotamon and Parapotamon A), semi-connected (Parapotamon B), semi-isolated (Pelsiopotamon), and isolated (Paleopotamon). The description of these types of water bodies are displayed in Table 2 with examples in Figure 7. Similar characterizations of water bodies by different types of hydrological connectivity have been made by Chovanec et al. (2005) and Funk et al. (2023).

Furthermore, spatial connectivity can be expressed as connectivity indices that account for landscape patterns or geographical distance (terrestrial and hydrological). For example, harmonic centrality, betweenness centrality (both weighed by landscape pattern), dPC flux or connector, and Moran's eigenvector maps (MEMs) can be used for such network analyses (Eros et al., 2024; Marchiori & Latora, 2000; Moran, 1948).

However, differences in species traits, such as life history, specialization to habitats, dispersal abilities (terrestrial or aquatic), and responses to disturbances (e.g., urbanization, floods, droughts, predation) are key factors in understanding vertebrate metacommunity patterns (Davis et al., 2016; Hamer et al., 2021; Péntek et al., 2017). As semiaquatic organisms, amphibians show different traits depending on the specific taxon and period of the year. For example, caudates like the Danube crested newt *Triturus dobrogicus* (KIRITZESCU 1903) and smooth newt *L. vulgaris* are more dependent on temporal water bodies than most anurans. Figure 4 gives an example of traits of selected species in terms of specialization to (aquatic) habitats on a local scale and terrestrial dispersal or migration on a regional scale. Below species traits of selected amphibian taxa out of the 21 native ones in Austria (14 anurans and 7 caudates) are described in more detail as reviewed by Kogoj (1997).

For instance, the distribution area of *T. dobrogicus* (Kogoj, 1997) is mostly restricted to floodplain areas with trenches, ponds, and marshes which are relatively deep and have open water areas with variable shading. During spawning season, they need macrophytes to attach their eggs (i.e., to single leaves), while their terrestrial habitats are under or in deadwood, roots, caves, or stones in immediate vicinity to their aquatic habitats. Their hibernation period, which is from late autumn till early spring, is mainly spent on land, despite aquatic hibernation being reported as well (Kogoj, 1997). *L. vulgaris* uses similar aquatic habitats with predominantly high sun exposure and aquatic plant richness that are surrounded by ruderal flora or forest. As terrestrial habitat, they use broadleaf and mixed forest, pits, as well as gardens, and hibernate below foliage, stones, and branches, as well as in their aquatic biotopes (Kogoj, 1997).

Anurans on the other hand tend to disperse over longer distances and have a broader spectrum of species traits, from generalists to specialists. For instance, the common toad *B. bufo* is a typical euryecious species (Kogoj, 1997) and can therefore tolerate a broad range of environmental conditions. Additionally, they can cope with the presence of fish due to a toxic compound (bufadienolides) in tadpoles (Hettyey et al., 2019; Kaczmarski et al., 2023). *B. bufo* predominantly uses large permanent water bodies close to forests with open water areas and intern structures to attach egg-strings during the spawning period (Kogoj, 1997). The summer period is mainly spent several hundred meters from their spawning habitats where they hunt nocturnally. They hibernate hidden below foliage, in holes or directly dug into the soil (Kogoj, 1997).

Lake frogs *Pelophylax ridibundus* (PALLAS 1771) and pool frogs *Pelophylax lessonae* (CAMERANO 1882) can form the hybrid *P.* kl. *esculentus* also called edible frog, which is difficult to differentiate from its parental species (Kogoj, 1997) anatomically as well as genetically. As a hybrid, *Pelophylax* kl. *esculentus* has a combination of traits from its parental species and therefore uses a broader spectrum of available habitats (Kogoj, 1997). Many traits of *P.* kl *esculentus* are similar to *P. ridibundus* which also mainly occupies sun exposed, deep, and large water bodies and uses aquatic and terrestrial plants for shelter during cold and windy periods or dig into the substrate of occupied water bodies (Kogoj, 1997). It is typical for the genus *Pelophylax* (water frogs) to usually stay within a few meters distance of the water during the summer and jump in to avoid danger. On the other hand, they can migrate long distances over land and thus are often one of the first to colonize newly formed water bodies but avoid vertical plant stands like reed belts and predominantly occupy habitats with horizontal aquatic vegetation. They are known to hibernate inside the ground of water bodies with a preference of slowly flowing waters as reported by Günther (1996a, b, c), whereas the hybrid species uses a broader range of habitats for hibernation like forests, gardens, or parks which seems to be population dependent (Kogoj, 1997)

Ranidae, like the common or grass frog *R. temporaria* and the agile frog *Rana dalmatina* (BONAPARTE 1840), are known to visit spawning habitats with open water, high sun exposure, and structural features like aquatic vegetation to attach their spawning clumps. Both use mainly standing but also slowly flowing spawning habitats with a minimum depth of 10 cm to 25 cm. However, *R. dalmatina* is known to be the first *Rana* species at the spawning sites, which is already in January or February (Kogoj, 1997). In the summer period, both species can be found in broadleaf forests with dense herbaceous layers. In Europe widely spread, *R. temporaria* can adapt to different environmental conditions with a preference to cold and humid conditions and hibernates in terrestrial, as well as in aquatic habitats, while *R. dalmatina* shows a tolerance against dry conditions and hence, can be found in dry sunny areas except of clear coniferous forests and hibernates most likely on land (Kogoj, 1997).

The common spadefoot *P. fuscus* predominantly uses large and deep permanent waters with structures like aquatic vegetation as spawning habitat and tolerates shading. Outside of the spawning period *P. fuscus* is known to dig into terrestrial areas with sandy to fine gravel soils between 200 m to 400 m away from spawning spots. This species is a nocturnal hunter, mainly in herbaceous vegetation. (Kogoj, 1997)

The fire-bellied toad *Bombina bombina* (L. 1758) is the most abundant species of the Danube floodplain (Cabela, 1990) and uses mid to large sized water bodies with open water zones and plant rich littoral zones which are not completely shaded as summer and spawning habitats that are permanent, at least until autumn (Kogoj, 1997). Newly formed water bodies are colonized quickly. Hibernation is undergone on land, in close vicinity or in the ground of the dried-out aquatic habitats, sometimes even inside water bodies. (Kogoj, 1997)

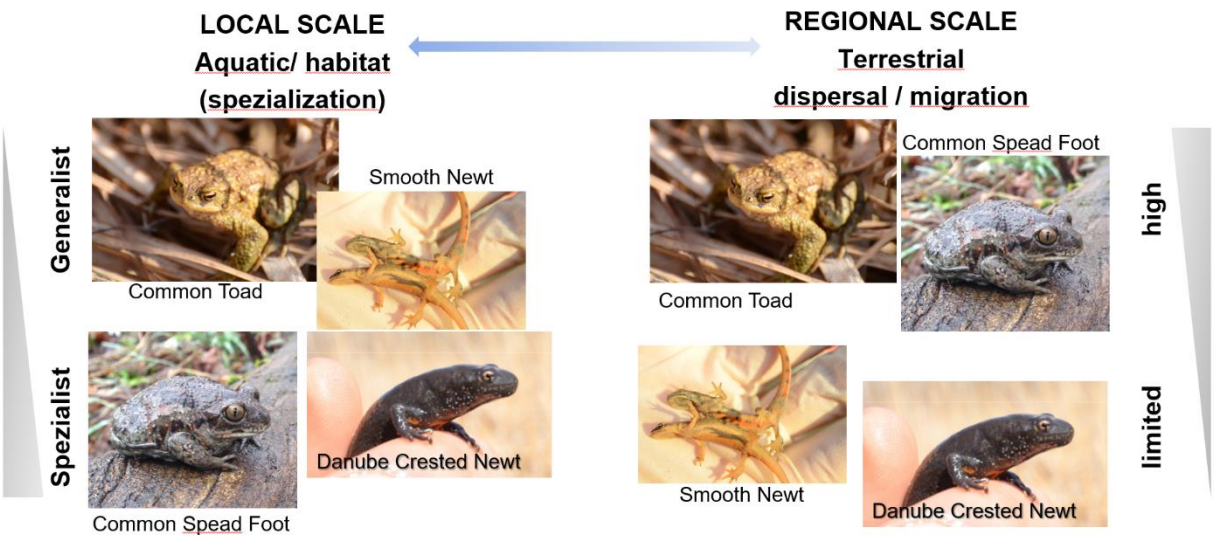

Figure 4: Example of species traits of selected species in terms of specialization to (aquatic) habitat on a local scale (left side) and terrestrial dispersal or migration ability on a regional scale (right side). Pictures by Peter Bader.

Traditional methods have been applied for several decades to investigate metacommunity patterns. Such physical surveys are rather time consuming and thus, cost-intensive compared to the innovative tool of environmental DNA (eDNA) (McKee et al., 2015; Olson et al., 2012). Hence, it is applied with increasing frequency, since it enables quick and non-invasive investigation. Environmental DNA consists of a complex mixture of fragmented DNA in the environment (e.g., water, soil, air, feces, etc.) stemming from different organismic origins (Taberlet et al., 2012). Taberlet et al. (2018) give an illustration of eDNA usage since the emergence of eDNA studies (Figure 5), as described below. It was first mentioned by Ogram et al. (1987) and applied to ancient sediment, but it became more common for investigating microorganisms post-millennial. Ficetola et al. (2008) performed the first application to aquatic organisms. However, in the last decade a strong increase in the usage of eDNA for ecological assessments on macro-organisms by utilization of water samples has occurred (Taberlet et al., 2012).

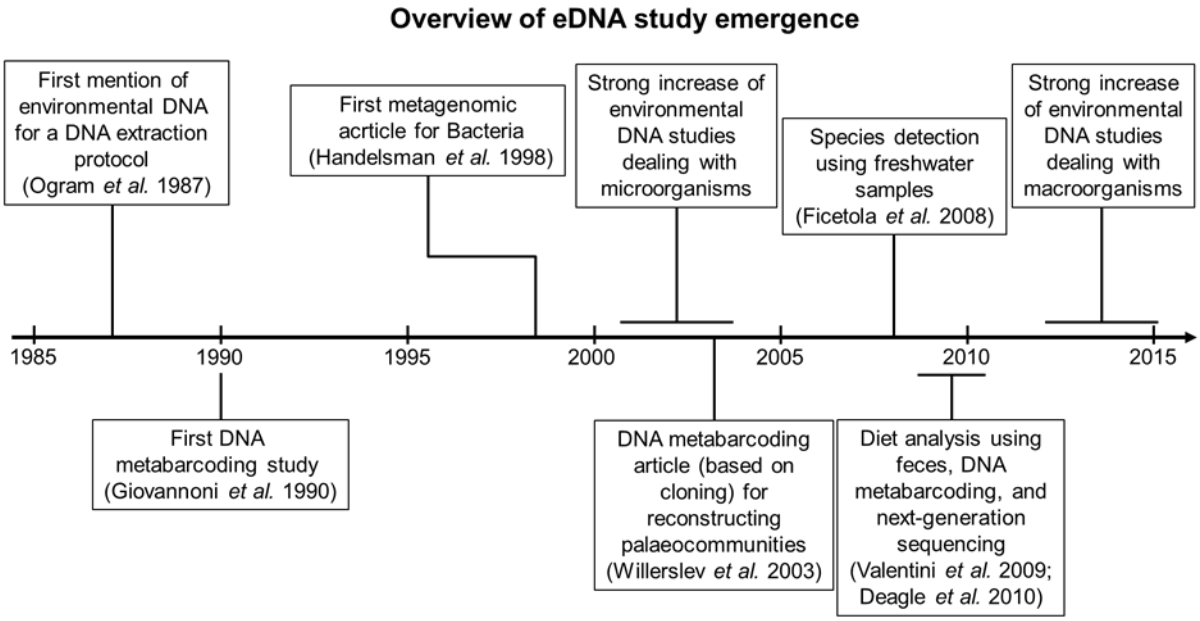

Figure 5: Illustration of eDNA usage in studies since their emergence. Adapted from Taberlet et al. (2018).

Several studies demonstrated that eDNA based surveys can be a more comprehensive tool to detect aquatic organisms like amphibians across various ecosystem types than traditional methods, specifically of rare and cryptic species (Dejean et al., 2012; Eiler et al., 2018; Li et al., 2021; Lopes et al., 2017; McKee et al., 2015; Olson et al., 2012; Thalinger et al., 2021; Thomsen et al., 2012). For example, Valentini et al. (2016) showed that at least four serial traditional investigations in the field are necessary to obtain the same success of amphibian detection compared to one single sampling using eDNA metabarcoding. However, many influencing factors must be considered when dealing with eDNA, because the sampled DNA is highly fragmented and present at very low concentrations (Deiner et al., 2021; Dejean et al., 2011); some of these factors are DNA degradation rates, eDNA transport, and differences in concentration (e.g., during or after amphibian spawning) (Lawson Handley et al., 2019; Pilliod et al., 2013; Thalinger et al., 2021).

Most preliminary eDNA studies in freshwater ecosystems were conducted in lotic (flowing) or lentic (standing) waters (e.g., Dejean et al., 2012; Lawson Handley et al., 2019; Li et al., 2021; Lopes et al., 2017; Pilliod et al., 2013; Thomsen et al., 2012), whereas highly dynamic systems like riverine floodplains are represented inadequately or not at all (Roni et al., 2019). Therefore, this study aims to assess (1) the detectability of amphibians in dynamic but altered riverine floodplains, (2) their spatial distribution within the floodplain, and (3) the relative importance of local abiotic and biotic environmental conditions, as well as spatial connectivity for amphibian metacommunity structure by using eDNA metabarcoding in waterbodies. To test these objectives, the free-flowing section of the Danube floodplain system between Vienna and Bratislava in Austria was used as an investigation area. This is one of the last remnants of fluvial ecosystems in the Upper Danube and in Central Europe (Preiner et al., 2018) with 67% of the native amphibian species in Austria (Nationalpark Donau-Auen, 2022).

Methods

The Danube is the most international and the second longest river in Europe, flowing through 10 ecoregions and 18 countries with more than 80 million people living in its basin (Hein et al., 2016; Tockner et al., 2009). It has a total length of over 2,800 km and drains an area of more than 800,000 km² (Hein et al., 2016; Reckendorfer et al., 2006; Tockner et al., 2009). The Danube lost more than 68% of its floodplains within the last 150 years, and up to 90% in the Upper Danube region (Hein et al., 2016; Tockner et al., 2009). The investigation area of this study is the “National Park Donau-Auen” in Lower Austria (Figure 6), a floodplain system in the free-flowing section of the Danube River in Austria (Europe) between Vienna (at stream kilometer 1,915.9) and the Slovakian border at Bratislava (at stream kilometer 1,881.2). There, it is a 9th order river according to Strahler with an average slope gradient of 0.45 ‰ (Reckendorfer et al., 2006; Strahler, 1957). With approximately 104,000 km² of drainage area at the national park (measuring station Wildungsmauer), low flow is 887 m³/s, and mean flow is 1,930 m³/s, while in the case of a 100 years flood discharge is up to 10,400 m³/s (noel.gv.at/wasserstand, 2022). Here, the Danube River historically exhibited a braided character but has been heavily regulated since 1875. Later, in 1990, restoration projects started to reconnect parts of the side channels (Funk et al., 2023; Reckendorfer et al., 2006). Currently, this enables water level fluctuations of up to seven meters during flood events, reshaping the area and creating habitats for a variety of biota (Chovanec et al., 2005; Shiel et al., 1998; Tickner et al., 2020).

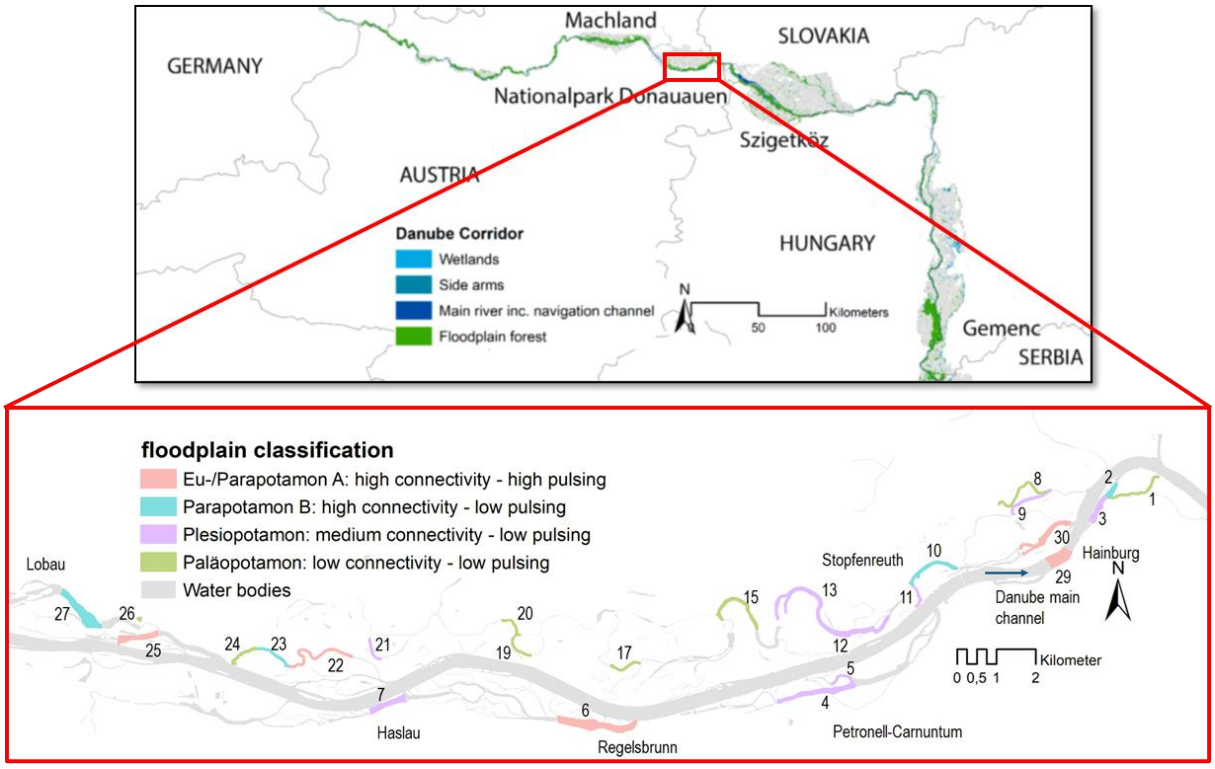

Figure 6: Study area in the National Park Donau-Auen in Lower Austria with the 26 sampled water bodies according to their classification. The flow direction of the Danube main channel is indicated by a blue arrow. Figure adapted from Funk (in Eros et al., 2024).

On an area of approximately 9,600 hectares, the study area in the “National Park Donau-Auen” consisted of roughly 65% floodplain forest, 15% meadows, and 20% water surface area (Nationalpark Donau-Auen, 2022). There, 26 water bodies were selected which were representative of their functional habitat (main river channel to isolated oxbows) and abundance according to their typological type (Eupotamon or Parapotamon A, Parapotamon B, Plesiopotamon, Paleopotamon; hereafter referred to as highly connected, semi-connected, semi-isolated, and isolated, respectively), and connectivity (low to high) (Chovanec et al., 2005; Eros et al., 2024; Funk et al., 2013, 2023; Hohensinner et al., 2011; Reckendorfer et al., 2006), as shown in Figure 7 and Table 2 (examples see Figure 8).

Table 2: Description of types of water bodies according to topology and hydrological connectivity, adapted from Amoros et al. (1982, 1987); and Hohensinner et al. (2011).

Habitat Type	Abbr.	synonymous used here	Definition	connected days (%)	amount sampled
Eupotamon or Parapotamon A	Eu-Para A	highly connected	main river channel (lotic); side arms (lotic): at low flow connected to the main channel at both ends; or highly dynamic backwaters (semi-lotic): abandoned braided channels/backwaters blocked upstream by bare gravel/sand deposits, with intact downstream connections; high pulsing	100-46	5
Parapotamon B	Para B	semi-connected	less dynamic backwaters (semi-lotic): abandoned braided channels/backwaters blocked upstream by vegetated deposits, with intact downstream connections; low pulsing	45-31	4
Plesiopotamon	Plesio	semi-isolated	isolated water bodies (lentic): temporary standing water ecosystems with frequent connections; or isolated water bodies (lentic): temporary standing water ecosystems with less frequent connections; low pulsing	30-1	9
Paleopotamon	Paleo	isolated	isolated water bodies (lentic): permanent standing water ecosystems; low pulsing	0	8

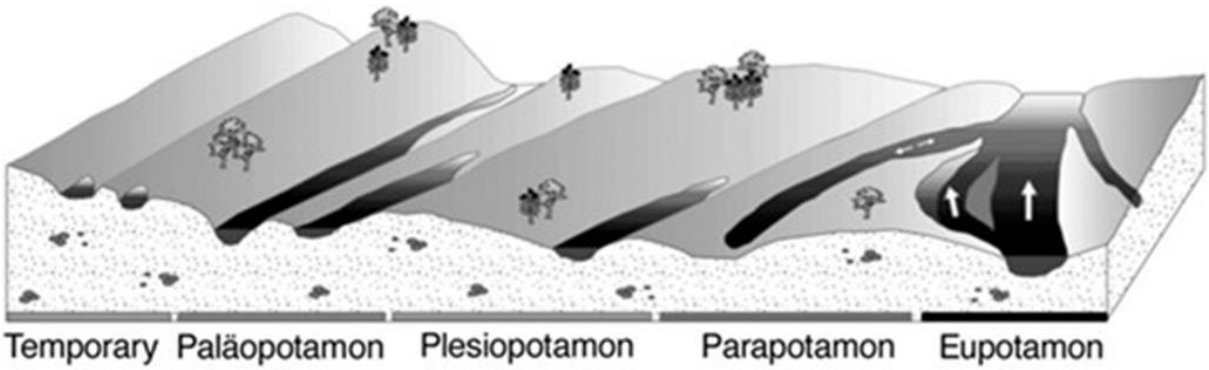

Figure 7: Types of water bodies according to typology after Amoros et al. (1982, 1987) and Hohensinner et al. (2011). Figure reprinted from Wahringer (in Graf et al., 2013).

In more detail, Chovanec et al. (2005) described 5 types of water bodies as follows. This includes (H1) sites with dynamic hydrology which are bidirectionally connected to the main channel and are characterized by high flow velocities, no macrophytes in the open water zones, open banks in the littoral zone, as well as sand and gravel as the dominating substratum. Secondly (H2), sites which have a one side connection to the main channel (at mean water levels), have few macrophytes, and still high proportions of sand and gravel. Third (H3), sites with no direct connection to the main channel at mean water levels have <20% of macrophytes at open water zones, and an increased degree of sedimentation. Forth (H4), are sites like H3 but with >20% of macrophytes in open water areas and high degrees of sedimentation. Fifth (H5), sites which are temporary pools that primarily depend on ground water levels, are dominated by macrophytes and terrestrial vegetation. Similarly, Funk et al. (2023) classified 5 groups of water bodies in floodplain systems by hydrological connectivity and indicators for their trophic status (nutrient content of phosphorous = P and nitrogen = N), sediment composition (based on nutrient and dissolved organic matter concentration), sediment grain size, organic content (and microbial activity), and macrophyte coverage. Group 1 is critical for flows in the system and is characterized by the highest overall connectivity, lowest fine-sediment content (organic and inorganic), lowest macrophyte cover, and highest nutrient loads (P and N). Group 2 has high total connectivity (directional transport is relevant) but is not critical for flow, has medium fine-sediment contents (on the banksides), low share of organic matter (OM), low macrophyte cover (on the banksides), and relatively high nutrient concentrations (P and N). Group 3 has high fine sediment (mainly river-based inorganic), and high unidirectional and network-independent connectivity. Group 4 has a lack of directional connectivity, reduced unidirectional and network-independent connectivity, high OM proportions in (fine) sediment, relatively high macrophyte coverage, and microbial byproduct-like matter (microbial activity gaining importance). Group 5 consists of overall disconnected sites with the highest proportion of OM in (fine) sediment, the highest macrophyte densities, has low nutrient content (P and N) in the water (especially at sites with higher isolation), but a high importance of microbial activity.

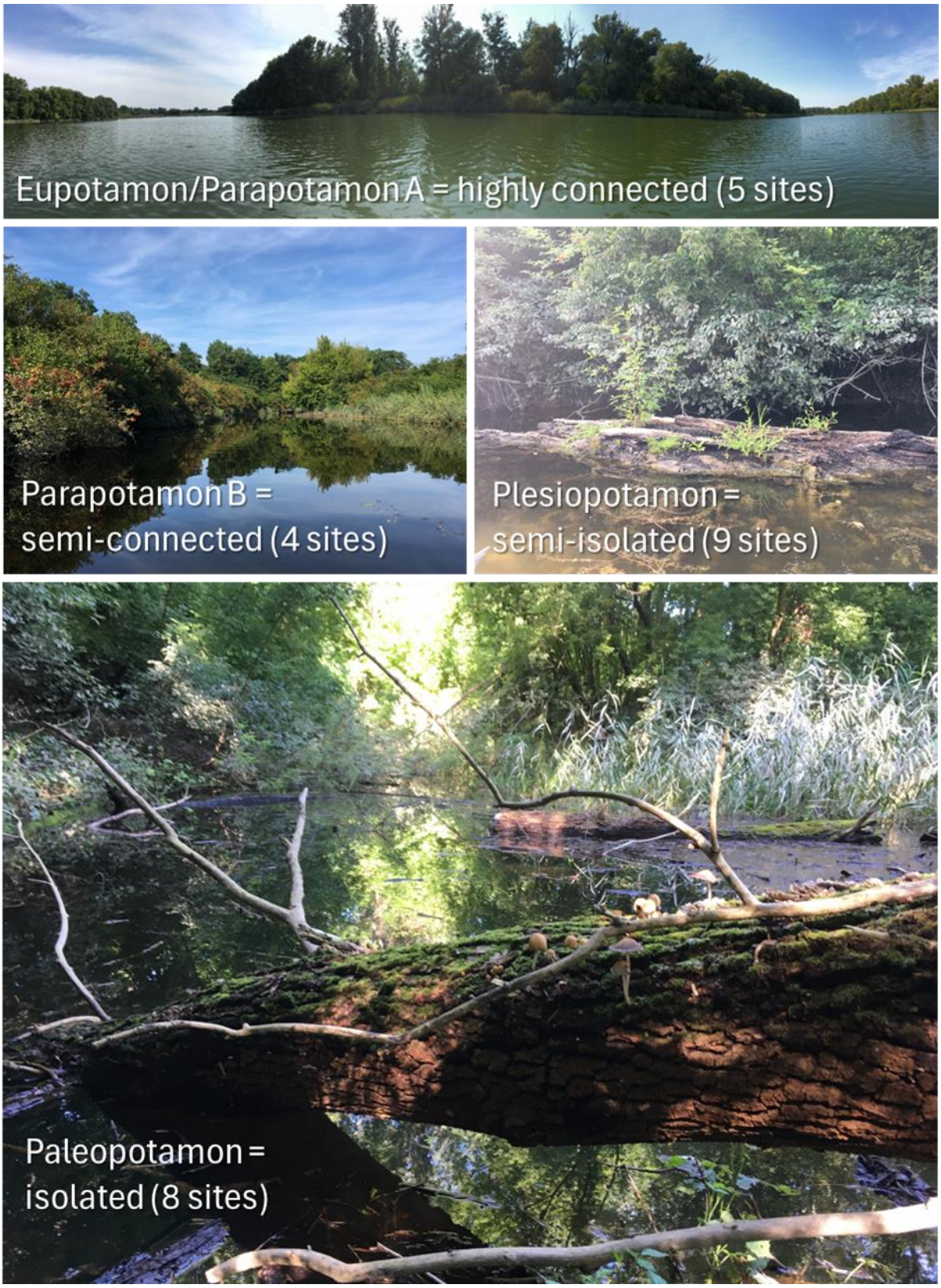

Figure 8: Example of the four selected typological water body types: Eu- or Parapotamon A = highly connected (top), Parapotamon B =semi-connected (middle left), Plesiopotamon = semi-isolated (middle right), and Paleopotamon = isolated (down). Images taken by Peter Bader.

The 26 selected sites were sampled twice (apart from three temporary sites due to insufficient water content), first in August 2021 (26 sites), approximately two weeks after a flood event, and second in September 2021 (23 sites).

In total 119 variables were inspected (see Table 3-Table 6). At each site, 40 environmental variables were recorded (see Table 3-Table 4, and compare Eros et al., 2024), in the field and in the laboratory. They include in field measurements of physio-chemistry parameters with portable sensors (i.e., water temperature in °C, pH, conductivity µS cm^{-1}, dissolved oxygen content in mg L^{-1} and percent; with a Hach-Lange HQ40; as well as water current velocity in cm sec^{-1} during mid and high water conditions using a water velocity meter), while visual estimation of in water habitat structures and substratum types (i.e., emergent vegetation, submerged vegetation, floating vegetation, floating/surface algae, open water habitat, and woody debris; as well as lithal, gravel, sand, and silt; all in percent, respectively) and bank structure composition (i.e., woody trees or large bushes, herbaceous vegetation, artificial, canopy cover, and fringing vegetation; all in percent).

Other environmental variables were calculated using GIS tools (i.e., water depth in meters and surface area in km² using a digital elevation model = DEM; area of the 5 types of land cover “agricultural”, “forest and other seminatural habitats”, “pasture”, “urban areas”, and “water or wetland” surrounding the water bodies within a buffer zone of 500 m in percent derived from the CORINE Land Cover 2018 database (European Environmental Agency, 2020, http://www.eea.europa.eu); and hydrological total, as well as upstream connectivity as mean number of days per year there is connection between a water body to the Danube main channel after Funk et al. (2013) and Reckendorfer et al. (2006) in days $year^{-1}$ and present $year^{-1}$; and influence of waterbodies from the flood event in 2021. Surface water levels needed for these calculations were provided from the Austrian River Authority (via Donau Österreichische Wasserstraßen GmbH).

The rest of the environmental variables were measured in the laboratory after taking water samples in the field (i.e., dry weight mg L^{-1}, ash weight mg L^{-1}, total dissolved phosphorus µg L^{-1}, colored dissolved organic matter = CDOM-humic Pt color mg L^{-1}, and chlorophyll-a µg L^{-1}). Quantification of total suspended solids (TSS) and particulate organic matter (POM) was done by filtering each water sample (between 300 and 2,000 ml) through pre-combusted GF/F filters (Whatman) (see Eros et al., 2024). The filters then were dried at 80°C for 24h and afterwards TSS was determined by measuring weight. For POM determination, the filters were weighed after ashing by 450°C for 4h. Using persulfate, total phosphorous (TP) was digested based on Eaton & Franson (2005) and analyzed according to ÖNORM EN ISO 15681-2. Measurement of CDOM was done after Cuthbert & del Giorgio (1992) by filtration of each water

sample through a Teflon membrane filters (0.45 µm) and measuring the light absorbance at 440 nm (Eros et al., 2024). Quantification of chlorophyll-a (chl-a) was done by filtering each water sample (between 300 and 2,000 ml) through pre-combusted GF/C filters (Whatman). Then, chl-a was extracted in cold acetone (90%) followed by measuring the absorption using a spectrophotometer according to Jeffrey & Humphrey (1975) and Lorenzen (1967) (see also Eros et al., 2024).

Later, 35 land or river based connectivity indices (connectivity variables) were calculated in the R environment (R Core Team, 2024) which are displayed in Table 5 (i.e., harmonic centrality and betweenness centrality of terrestrial and river based Euclidean distance, dPC flux and connector indices of terrestrial and river based distance with thresholds of 100 m/1.000 m/10.000 m each, as well as Moran's Eigenvector Maps further on referred to as MEMs) (Borgatti, 2005; Eros et al., 2024; Funk et al., 2023; Marchiori & Latora, 2000; Marra et al., 2014; Moran, 1948; Rochat, 2009; Saura & Rubio, 2010). Additionally, eDNA sequence-data from a total of 44 detected fish taxa were used as biotic independent variables (see Table 6).

Table 3: Overview-table of environmental variables including units, description, delineation and corresponding literature (part 1 of 2).

Environmental Variables (n=40)	Units	Description	Delineation	Literature
area	km²	water surface area	GIS tools + surface water levels	Eros et al., 2024
depth	m	mean water depth	GIS tools using DTM	
emergent	%	surface composition within a water body	visual estimation during sampling	Eros et al., 2024
submerged				
floating vegetation				
surface/floating algae				
open water habitat				
woody debris				
woody (trees, large bushes)	%	bank structure surrounding a water body	visual estimation during sampling	Eros et al., 2024
herbaceous				
artificial (concrete, rip-rap)				
canopy cover				
fringing vegetation				
lithal	%	substrate class within a water body	visual estimation during sampling	Eros et al., 2024
gravel				
sand				
silt				
sediment depth	cm	depth of the layer of fine sediment	measured in the field with a profile rod	-
velocity (HSW)	cm/s	current velocity during high water conditions	water velocity meter	Eros et al., 2024
velocity (MW)	cm/s	current velocity during mid water conditions		
oxigen	%	dissolved oxygen	measured during sampling using a portable sensor (Hach-Lange HQ40)	Eros et al., 2024
oxigen	mg/l	dissolved oxygen		
conductivity	µS/cm	conductivity		
pH	-	pH-value		
temperature	°C	water surface temperature		
dry weight	mg/l	total suspended solids (TSS)	measured in the laboratory from water samples taken in the field	Eros et al., 2024
ash weight	mg/l	particulate organic matter (POM)		Eros et al., 2024
TP	µg/l	total dissolved phosphorus		Eaton & Franson, 2005; Eros et al., 2024
CDOM-humic(Ptcolour)	mg/l	coloured dissolved organic matter (humic colour)		Cuthbert & del Giorgio, 1992; Eros et al., 2024
chlorophyll-a	µg/l	chlorophyll a concentration		Loreznzen 1967; Jeffrey 1975; Eros et al., 2024
agricultural	%	land-use cover within a 500m wide buffer area around the water body	GIS tools + obtained from the CORINE Land Cover 2018 database	Eros et al., 2024; European Environmental Agency, 2020, (http://www.eea.europa.eu)
forest and seminatural				
pasture				
urban				
water and wetland				

Table 4: Continuation of the overview-table of environmental variables including units, description, delineation and corresponding literature (part 2 of 2).

Environmental Variables (n=40)	**Units**	**Description**	**Delineation**	**Literature**
total connectivity	days/ year	mean number of days in a year a waterbody is connected to the main Danube channel	GIS tools + surface water levels (derived from Austrian River Authority: via Donau, Österreichische Wasserstraßen GmbH)	Eros et al., 2024; Funk et al., 2013; Reckendorfer et al., 2006
upstream connectivity	days/ year	mean number of days in a year a waterbody is connected to the main Danube channel upstream		
total connectivity	days in %	% of mean number of days in a year a waterbody is connected to the main Danube channel		
upstream connectivity	days in %	% of mean number of days in a year a waterbody is connected to the main Danube channel upstream		
Floodevent2021	0 or 1	0 = not affected; 1 = affected		

Table 5: Overview-table of delineated connectivity variables, including descriptions, corresponding literature, and meaning of abbreviations (below).

Connectivity Indices (n=35)				Description	Literature
eu		hc		euclidean distance based harmonic centrality	Funk et al., 2023; Marchiori & Latora, 2000; Rochat, 2009; Borgatti, 2005
		bc		euclidean distance based betweenness centrality	Funk et al., 2023; Borgatti, 2005; Marra et al., 2014
rb		hc		river based harmonic centrality	Funk et al., 2023; Marchiori & Latora, 2000; Rochat, 2009; Borgatti, 2005
		bc		river based betweenness centrality	Funk et al., 2023; Borgatti, 2005; Marra et al., 2014
100	eu	flux	dPC	euclidean distance based dPC flux with a threshold of 100m	Saura & Rubio, 2010
1000				euclidean distance based dPC flux with a threshold of 1000m	
10000				euclidean distance based dPC flux with a threshold of 10000m	
100	rb			river based dPC flux with a threshold of 100m	
1000				river based dPC flux with a threshold of 1000m	
10000				river based dPC flux with a threshold of 10000m	
100	eu	connector		euclidean distance based dPC connector with a threshold of 100m	
1000				euclidean distance based dPC connector with a threshold of 1000m	
10000				euclidean distance based dPC connector with a threshold of 10000m	
100	rb			river based dPC connector with a threshold of 100m	
1000				river based dPC connector with a threshold of 1000m	
10000				river based dPC connector with a threshold of 10000m	
MEM1 - MEM19				Moran's Eigenvector Map(s) (MEM) values (1 ≙ large to 19 ≙ small scale pattern of community composition)	Eros et al., 2024; Moran, 1948

Abbreviation	Meaning
eu	euclidean distance (terrestrial connection)
rb	river based (euclidean distance along the hydrological connection)
hc	harmonic centrality
bc	betweenness centrality
dPC	percentage of the variation in PC (= probability of connectivity)
flux	includes the surface area of water bodies
connector	accounts for the importance of water the water body as stepping stone
100, 1000, 10000	threshold distance in meters
MEM	Moran's Eigenvector Map

Table 6: Overview of biotic variables (fishes) delineated by eDNA metabarcoding. The table includes taxa names, English names, total number of water bodies with detections. Taxa with negative effects on amphibians (after Laufer & Wollenzin, 2011) are highlighted in red and by an X on the right-hand side.

Taxon (Biotic Variables; n=44)	**English Name**	**Amount of Water Bodies with Detection**	**Negative Impact on Amphibians (Laufer & Wollenzin, 2011)**
Abramis brama	Bream	21	X
Alburnoides bipunctatus	Schneider	4	
Alburnus alburnus	Bleak	22	X
Ameiurus melas	Black bullhead	1	
Anguilla anguilla	European eel	1	X
Aspius aspius	Asp	24	
Babka gymnotrachelus	Racer goby	20	
Ballerus ballerus	Blue bream	1	
Barbatula barbatula	Stone loach	2	
Barbus barbus	Barbel	9	
B.sapa-B.bjoernkna-V.vimba	***species complex*** (White-eye bream, Silver bream, Vimba bream)	17	
Carassius gibelio	Prussian carp	21	
Cobitis elongatoides	Spiny loach	1	
Cottus gobio	European bullhead	6	
Cyprinidae - Complexe 1	***species complex*** (Carp family)	6	
Cyprinus carpio	Carp	19	X
Esox lucius	Pike	22	X
Gasterosteus aculeatus	Three-spined stickleback	8	X
G.gobio-Ramonogobio sp.	***species complex*** (Gudgeon complex)	4	X
Gymnocephalus sp.	***species complex*** (Ruffe complex)	16	X
Hypophthalmichthys molitrix	Silver carp	8	
Hypophthalmichthys nobilis	Bighead carp	4	
Lepomis gibbosus	Pumpkinseed	25	X
Leucaspius delineatus	Sunbleak	2	X
L.idus-L.leuciscus-P.cultratus	***species complex*** (Ide, Eurasian daces, Sichel)	22	
Misgurnus fossilis	Weatherfish	3	
Neogobius melanostomus	Round goby	13	
Perca fluviatilis	European perch	24	X
Phoxinus sp.	Minnow sp.	1	X
Ponticola kessleri	Bighead goby	12	
Proterorhinus semilunaris	Tubenose goby	22	
Pseudorasbora parva	Stone moroko	11	
Rutilus virgo-R. pigus	***species complex*** (Cactus Roach, Pigo)	17	
Rhodeus amarus	European bitterling	22	X
Rutilus rutilus	Roach	22	X
Sander sp.	***species complex*** (Pike-perch)	21	X
Scardinius erythrophthalmus	Rudd	19	X
Silurus glanis	European catfish	17	X
Squalius cephalus	Chub	18	X
Thymallus thymallus	Grayling	2	X
Tinca tinca	Tench	18	X
Umbra krameri	European mudminnow	1	
Zingel streber	Streber	2	
Zingel zingel	Zingel	4	

Simultaneously to the environmental survey, eDNA samples were taken according to Eros et al. (2024; see also Pont et al., 2019, 2022). This included in-situ filtration of approximately 28 Liters on average (from 6.6 up to 94 Liters, dependent turbidity) at roughly 20 cm water depth per sample using a peristaltic pump, VigiDNA cross-filtration capsules (with 0.45 µm pore size; SPYGEN, Le Bourget-du-Lac, France), and sterile single-use equipment ,like plastic tubes with protection filter caps which were attached to a rod, gloves, and more (see Figure 9). Depending on the water level, size, and accessibility of the water body, this was done by carefully and slowly wading through the water body or from a boat. To avoid eDNA cross-contamination among sampling sites, one filter was used per site and session and the filtration equipment (i.e., plastic tube with protection filter cap attached to a rod) was positioned upstream from the filtration area (see Figure 9 left and middle) and showing directly towards the direction of movement (Eros et al., 2024).

Figure 9: Used equipment for taking eDNA samples, like peristaltic pump (left, middle), cross-filtration capsules (left, middle, right), and single-use equipment (right) like plastic tubes, gloves, etc. Images taken by Peter Bader.

After filtering the required volume of water (or until the cross-filtration capsule was clogged) 80 ml of conservation buffer CL1 (SPYGEN) were added for DNA-preservation, to avoid eDNA degradation. Afterwards, the samples were sent to the SPYGEN laboratory (specialized in eDNA analyses) in France for further processing (SPYGEN). There, they conducted DNA extraction, amplification (using universal *Teleo* primers for fish and *Batra* primers for amphibian detection, 12S mtDNA, 50 PCR cycles, 12 PCR replicates per sample, and 3 negative controls), purification, and sequencing according to Eros et al. (2024) twice for quality assurance. Taxonomic assignment via reference databases (for amphibians: *SPYGEN*, *AUTRICHE*, and *EMBL 142*; for fishes: *SPYGEN* and *DANUBE 2019*) was done according to Pont et al. (2022), resulting in two original versions of data sets (v1 and v2). For a better taxa assignment, a local database of amphibian DNA sequences was constructed beforehand by supplying available tissue of certain amphibian species to the SPYGEN laboratory (SPYGEN). Thus, results represent semi-quantitative data.

Data preparation and statistical analysis

The two gained data sets (v1 and v2) of sequence reads were prepared in MS Excel (Excel Microsoft 365) as follows. Redundant information was uniformed or deleted. For example, in one data set (v1) there were three similar taxa of *Pelophylax* (*P. ridibundus*, *P.* kl. *esculentus*/ *ridibundus*, and *P.* Complex 1). Additionally, there was one high level (family) taxon Ranidae and one low level taxon (species) *R. temporaria*, while the other data set (v2) contained detections of *R. dalmatina*. Therefore, only one taxon of each was kept, named *Pelophylax* sp and *Rana* sp. respectively. Furthermore, redundant detections were deleted and only the greatest number of sequences reads were adapted per taxon and site in each data set. For comparison, both original data sets (v1 and v2) were combined into a single data set (pooled) following the same procedure.

According to these three prepared data sets (v1, v2, and pooled) abiotic (environmental and connectivity) and biotic (fish) variables were matched respectively. These obtained data sets were then imported into R (R Core Team, 2024; version 4.4.1) and checked for normal distribution (by creating histograms using the function “geom_histogram” as well as “ggplot” in the package *ggplot2*, version 3.5.1; as well as the function “shapiro.test” in the package *stats*, version 4.4.1, respectively) and skewness (using the function “skewness” in the package *moments*, version 0.14.1). After testing for the best fitting transformations (using the functions “transformTukey”, “log”, and “decostand” in the packages *rcompanion*, version 2.4.36, *base*, version 4.4.1, and *vegan*, version 2.6-6.1, respectively; and histograms), abiotic data (environmental and connectivity variables with non-normal distribution) was Tukey transformed (using the function “transformTukey”) while biotic data (for fish and amphibian) was Hellinger transformed (using the function “decostand”). Next, basic statistical analyses like Shannon diversity index (using the “diversity” function in the package *vegan*, version 2.6-6.1) and Evenness (using the function “evenness” in the package *BAT*, version 2.9.6) were performed. This was done after transformation since the amphibian data consists out of DNA sequence reads and not real abundance data. The local contribution to beta diversity (LCBD) was calculated using the functions “alpha”, “beta”, and “LCBD.comp” in the *BAT* package (version 2.9.6).

Then, all datasets were condensed into having sites with only amphibian detections (sites without amphibian detections were deleted) and MEMs were calculated (using the “dbmem” function in the package *adespatial*, version 0.3-23; see also Borcard et al., 2011) and added to the abiotic data sets respectively. Abiotic data was matched according to the condensed datasets and split into environmental variables (environmental model), connectivity indices (connectivity model), and biotic variables (fish model) for further and multivariate analyses (i.e.,

RDA, ANOVA, and variance partitioning) to examine the relationship between environmental conditions and metacommunity composition. Thus, separate forward selection of best fitting models using a permutation-based test (function “ordistep”, 200 runs) was performed and for RDAs the function “rda” both in the package *vegan* (version 2.6-6.1) (Eros et al., 2024), as well as the functions “otu_table”, “taxa_table”, “sample_data”, and “phyloseq” in the package *phyloseq* (version 1.48.0) were used for further data exploration, respectively. ANOVAs were calculated using the “anova.cca” function (package *vegan*, version 2.6-6.1). Only significant variables (p-value < 0.05) related to amphibian community variability were kept for further analyses and the final models (Eros et al., 2024). After testing all three dataset versions (v1, v2, and pooled) and all models (environmental, connectivity, fish) separately, only the combined (pooled) dataset was used due to the best fitting and greatest statistical significance. For plotting the figures of multivariate analyses like RDA plots (using the functions “ord_explore”, “tax_transform”, “ord_calc”, and “ord_plot” in the package *microViz*, version 0.12.3), only variables with statistical significance (p-value < 0.05) were used for better visualization. With the selected (pooled) models, variance partitioning procedure (using the “varpart” function in the *vegan* package, version 2.6-6.1) was performed (Borcard et al., 2011) to evaluate the pure and shared effects of environmental, connectivity, and biotic (fish) variable groups on amphibian diversity (Eros et al., 2024). The “plot” function (in the *base* package, version 4.4.1) was used for visualization of the variance partitioning.

Results

Overall, amphibian eDNA was detected at 20 out of 26 sampled sites (Table 7), stemming from two species of caudates (*T. dobrogicus* and *L. vulgaris*) and five anuran taxa (*B. bombina*; *B. bufo*; *P. fuscus*; *P.* kl. *esculentus* and/or *P. ridibundus* hereinafter referred to as *Pelophylax* sp.; as well as *R. temporaria* and/or *R. dalmatina* hereinafter referred to as *Rana* sp.) in total. The eDNA of *Pelophylax* sp. was detected (several times also visually) in 18 of the 20 occupied sites making it the most abundant taxon, followed by *P. fuscus* with 6 detections. *B. bufo,* and *L. vulgaris* were detected at 4 sites followed by *Rana* sp. with 3 detections, then by *B. bombina* with 2 (once also visually) and *T. dobrogicus* with a single detection.

In August, amphibian eDNA was detected at 14 sites of the 26 sampled sites (see Table 7 and Figure 10). Six of them showed recordings of amphibians merely in August, however, three of which are temporary water bodies and were not sampled in September due to insufficient water content. Likewise, in September, 14 sites contained amphibian eDNA out of the 23 sampled sites with merely 6 recordings (Table 7 and Figure 10).

Table 7-Table 8 show the occurrence of amphibians at different sites and months. At the site with the highest species richness (only sampled in August) eDNA of five different amphibian taxa were recorded (dominated by *Pelophylax* sp., *L. vulgaris*, *P. fuscus*, *Rana* sp., and *T. dobrogicus*, respectively). One site showed traces of four different amphibian taxa (*Pelophylax* sp., *P. fuscus*, *L. vulgaris*, and *Rana* sp.). Three sites contained eDNA from three amphibian taxa (all of them with *Pelophylax* sp. and *P. fuscus*; two additionally with *B. bombina*, and one with *L. vulgaris* as third taxon), whereas one was only sampled in August. At five sites there was eDNA of two amphibian genera detected (all of them with *Pelophylax* sp., two with *B. bufo*, and the other three with either *P. fulscus*, *L. vulgaris*, or *Rana* sp.), one of them only sampled in August. At the other ten sites only one amphibian taxon was detected (one with *L. vulgaris*, two with *B. bufo*, and eight with *Pelophylax* sp.).

Table 7: Overview-table of amphibian occurrences per site and month. Cells highlighted in green show detections in both months, blue highlighted cells show detections only in August 2021 (and the three sites only sampled in August 2021), and red highlighted cells show detections only in September 2021. Empty cells in the column 'month' refer to sites with at least one detected taxon in both months.

site-ID	month	site name	*Bombina bombina*	*Bufo bufo*	*Lissotriton vulgaris*	*Pelobates fuscus*	*Pelophylax sp. (P. kl esculentus, P. ridibundus)*	*Rana sp. (R. dalmatina, R. temporaria)*	*Triturus dobrogicus*	sum of taxa detections per site
1		Roethelstein isoliert			x		x			2
2	August	Roethelstein flussab					x			1
4		Petronell					x	x		2
5	August	Petronell isoliert		x			x			2
7	August	Haslau					x			1
8	September	Tiergartenarm isoliert				x	x			2
9		Tiergartenarm					x			1
10	September	Karpfenarm		x						1
11	August	Karpfenarm isoliert			x	x	x			3
12		Rosskopfarm innen	x			x	x			3
13	September	Rosskopfarm außen					x			1
19	August	Fadenbach innen			x	x	x	x	x	5
20		Fadenbach außen			x	x	x	x		4
21		Fadenbach bei Orth					x			1
22	September	Kleine Binn					x			1
23	August	Rohrwoerth flussab		x						1
24		Rohrwoerth flussauf		x			x			2
26		Lobau Grundwassertuempel	x			x	x			3
27	September	Schoenauer Wasser					x			1
30	September	Spittelauer Arm					x			1
count of detections per taxon			2	4	4	6	18	3	1	

Figure 10 shows the percentage of detected amphibian eDNA sequences per site and month according to water body types (connectivity). In highly connected water bodies (Eupotamon or Parapotamon A) only *Pelophylax* sp. was detected. In semi-connected water bodies (Parapotamon B) *Pelophylax* sp. and *B. bufo* were detected uniquely in two sites each. At semi-isolated sites (Plesiopotamon), the genus *Pelophylax* was uniquely detected eight times, and twice additionally with one other taxon and twice with two other taxa. In one of the isolated sites, *Pelophylax* sp. was the only detected genus. In three of the isolated sites, one other taxon was detected next to *Pelophylax* sp. (once *P. fuscus* and twice *B. bufo*). Two isolated sites contained eDNA of 3 different taxa and at one site the highest number of taxa (5) was detected, from *T. dobrogicus*, *L. vulgaris*, *P. fuscus*, *Rana* sp. and the genus *Pelophylax*.

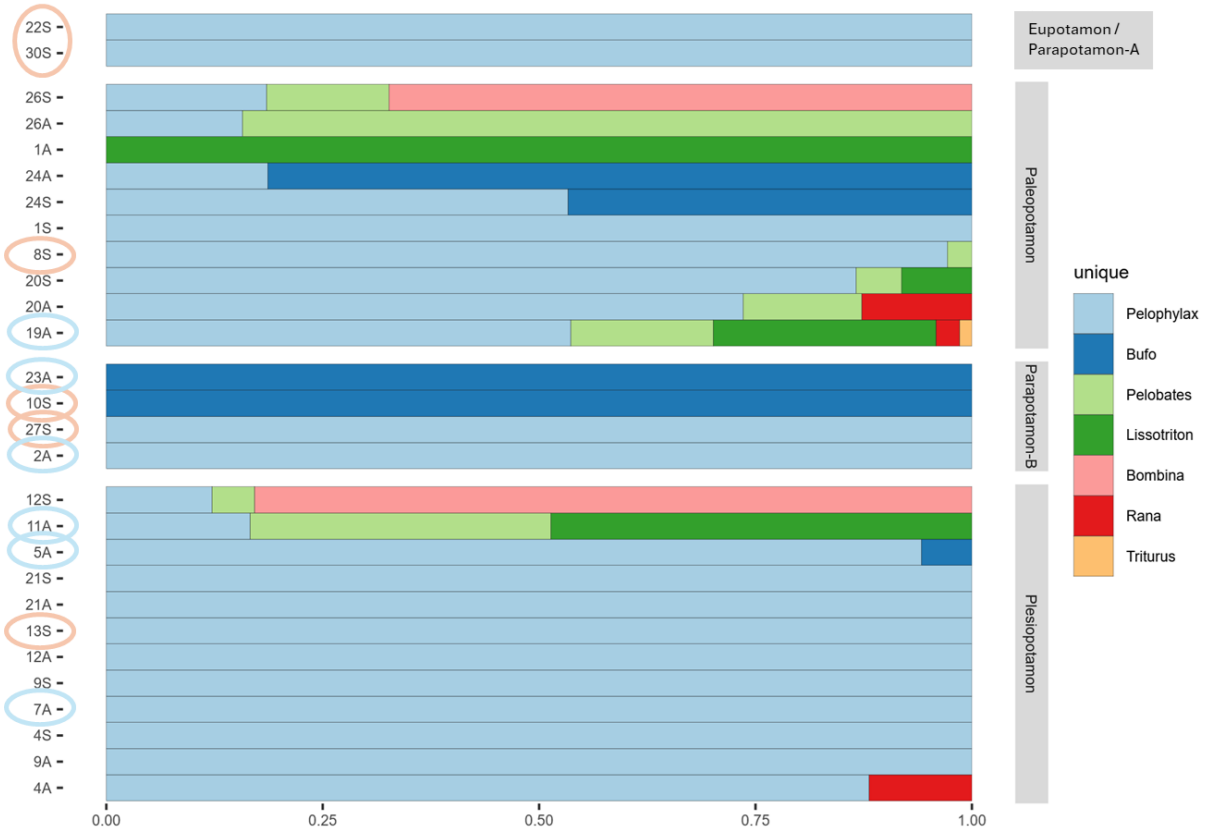

Figure 10: Percentages of detected eDNA sequences per site and month with amphibian detections according to water body type (connectivity). From top to down: Eupotamon or Parapotamon A (highly connected), Paleopotamon (isolated), Parapotamon-B (semi-connected), Plesiopotamon (semi-isolated). On the left side, the site codes are displayed, whereas A stands for sites sampled in August 2021 and S stands for sites sampled in September 2021. The circled site codes highlight sites with detections merely in that month (blue for August and red for September).

Table 8 shows an overview of the calculated Shannon indices, evenness indices, and LCBD (local contributions to beta diversity) values per site and month with amphibian detections. The lowest Evenness value out of the twelve calculated ones was 0.06, followed by 0.12, and 0.24, while four were between 0.30 and 0.50, and the other five were above 0.50 with the highest value being 0.93. The highest Shannon index was 1.14, followed by 1.02. All other Shannon values were below 1.00. The alpha values of the LCBD reflect the results of the Shannon indices. Four sites show an LCBD beta value higher than 0.08, and four sites had a value between 0.04 and 0.08. The other twelve sites showed a lower value than 0.04. Likewise, there were the same four sites with high LCBD repl (replacement) values higher than 0.1, but only three values between 0.04 and 0.1, whereas the rest (21) had values below 0.04. The highest LCBD rich (richness difference) values above 0.1 were at four different sites compared to the LCBD beta and repl values from before. Six of the LCBD rich values were between 0.03 and 0.1, whereas the rest of the sites (18) had a value below 0.02.

Table 8: Overview-table of LCBD (local contribution to beta diversity) values, Shannon index, and evenness index per site (split in month and type of water body). Cells with Shannon and LCBD indices as well as number of detected taxa are highlighted in green according to the values (the higher the value the stronger the color).

ID	month	type of water body	connectivity	site name	Shannon index	alpha Mean (LCBD)	beta LCBD	repl LCBD	rich LCBD	Evenness index	number of detected taxa
1A	August	Paleopotamon	isolated	Roethelstein isoliert	0.00	1.00	0.105	0.178	0.004	-	1
1S	September	Paleopotamon	isolated	Roethelstein isoliert	0.00	1.00	0.010	0.008	0.012	-	1
2A	August	Parapotamon-B	semi-connected	Roethelstein flussab	0.00	1.00	0.010	0.009	0.011	-	1
4A	August	Plesiopotamon	semi-isolated	Petronell	0.36	1.23	0.024	0.014	0.037	0.24	2
4S	September	Plesiopotamon	semi-isolated	Petronell	0.00	1.00	0.010	0.008	0.013	-	1
5A	August	Plesiopotamon	semi-isolated	Petronell isoliert	0.22	1.06	0.012	0.008	0.017	0.12	2
7A	August	Plesiopotamon	semi-isolated	Haslau	0.00	1.00	0.009	0.005	0.014	-	1
8S	September	Paleopotamon	isolated	Tiergartenarm isoliert	0.13	1.02	0.010	0.008	0.014	0.06	2
9A	August	Plesiopotamon	semi-isolated	Tiergartenarm	0.00	1.00	0.010	0.009	0.012	-	1
9S	September	Plesiopotamon	semi-isolated	Tiergartenarm	0.00	1.00	0.010	0.007	0.013	-	1
10S	September	Parapotamon-B	semi-connected	Karpfenarm	0.00	1.00	0.097	0.166	0.002	-	1
11A	August	Plesiopotamon	semi-isolated	Karpfenarm isoliert	1.02	2.78	0.069	0.019	0.138	0.79	3
12A	August	Plesiopotamon	semi-isolated	Rosskopfarm innen	0.00	1.00	0.010	0.008	0.012	-	1
12S	September	Plesiopotamon	semi-isolated	Rosskopfarm innen	0.56	1.34	0.086	0.121	0.038	0.48	3
13S	September	Plesiopotamon	semi-isolated	Rosskopfarm außen	0.00	1.00	0.010	0.008	0.013	-	1
19A	August	Paleopotamon	isolated	Fadenbach innen	1.14	2.84	0.060	0.000	0.142	0.74	5
20A	August	Paleopotamon	isolated	Fadenbach außen	0.76	1.99	0.043	0.014	0.083	0.59	3
20S	September	Paleopotamon	isolated	Fadenbach außen	0.48	1.18	0.018	0.009	0.032	0.46	3
21A	August	Plesiopotamon	semi-isolated	Fadenbach bei Orth	0.00	1.00	0.010	0.008	0.013	-	1
21S	September	Plesiopotamon	semi-isolated	Fadenbach bei Orth	0.00	1.00	0.010	0.008	0.012	-	1
22S	September	Eu-Parapotamon-A	highly connected	Kleine Binn	0.00	1.00	0.009	0.007	0.013	-	1
23A	August	Parapotamon-B	semi-connected	Rohrwoerth flussab	0.00	1.00	0.097	0.166	0.003	-	1
24A	August	Paleopotamon	isolated	Rohrwoerth flussauf	0.48	1.68	0.062	0.058	0.067	0.37	2
24S	September	Paleopotamon	isolated	Rohrwoerth flussauf	0.69	2.00	0.047	0.008	0.101	0.93	2
26A	August	Paleopotamon	isolated	Lobau Grundwassertuempel	0.44	1.49	0.074	0.090	0.052	0.31	2
26S	September	Paleopotamon	isolated	Lobau Grundwassertuempel	0.86	2.29	0.068	0.040	0.107	0.65	3
27S	September	Parapotamon-B	semi-connected	Schoenauer Wasser	0.00	1.00	0.009	0.007	0.013	-	1
30S	September	Eu-Parapotamon-A	highly connected	Spittelauer Arm	0.00	1.00	0.010	0.010	0.011	-	1

All calculated ANOVA results, including the p-values of the different RDA analysis of final models (of the pooled data sets) are displayed in Table 9-Table 11. RDA analysis of environmental parameters (Figure 11) explains up to 30% of variance on the first axis and 13% on the second axis. In further detail, it showed a positive correlation (Table 9) of aquatic algal biomass (chlorophyll a concentration; p-value = 0.013) with *B. bombina*, *P. fuscus*, and *L. vulgaris* occurrence and a negative correlation with water depth (p-value = 0.001), as well as surface area (p-value = 0.031) with these three species. Furthermore, it showed a negative correlation of silt (p-value = 0.008) with the taxon *Pelophylax* and a negative correlation of the percentage of herbaceous plants area in the surrounding of water bodies with *B. bufo* (p-value = 0.063).

Table 9: ANOVA result of the (pooled) environmental model (RDA) showing only statistically significant variables.

Variable	***Df***	***Variance***	***F***	***p***	***Level of significance***
Depth (m)	1	0.064319	6.5072	0.001	*** = ≤ 0.001
Silt (%)	1	0.050396	5.0987	0.008	** = 0.001-0.01
Chlorophyl a	1	0.039933	4.0401	0.013	* = 0.01-0.05
Area (km²)	1	0.037659	3.8101	0.031	* = 0.01-0.05
Herbaceous (%)	1	0.025710	2.6012	0.063	+ = 0.1-0.05
Residual	22	0.217451			

Table 10: ANOVA result of the (pooled) connectivity model (RDA) showing only statistically significant variables.

Variable	***Df***	***Variance***	***F***	***p***	***Level of significance***
MEM3	1	0.063066	6.1621	0.003	** = 0.001-0.01
Eu1000 flux	1	0.038607	3.7722	0.015	* = 0.01-0.05
MEM10	1	0.036759	3.5916	0.020	* = 0.01-0.05
MEM5	1	0.035834	3.5013	0.023	* = 0.01-0.05
MEM22	1	0.036039	3.5213	0.025	* = 0.01-0.05
Residual	22	0.225162			

Table 11: ANOVA result of the (pooled) biotic (fish) model (RDA) showing only statistically significant variables.

Variable	***Df***	***Variance***	***F***	***p***	***Level of significance***
Cobitis elongatoides	1	0.051785	5.5037	0.002	** = 0.001-0.01
Perca fluviatilis	1	0.040660	4.3213	0.004	** = 0.001-0.01
Gymnocephalus sp.	1	0.044845	4.7661	0.008	** = 0.001-0.01
Alburnus alburnus	1	0.044679	4.7484	0.011	* = 0.01-0.05
Scardinius erythrophthalmus	1	0.027793	2.9538	0.033	* = 0.01-0.05
Rutilus rutilus	1	0.028113	2.9878	0.040	* = 0.01-0.05
Residual	21	0.197593			

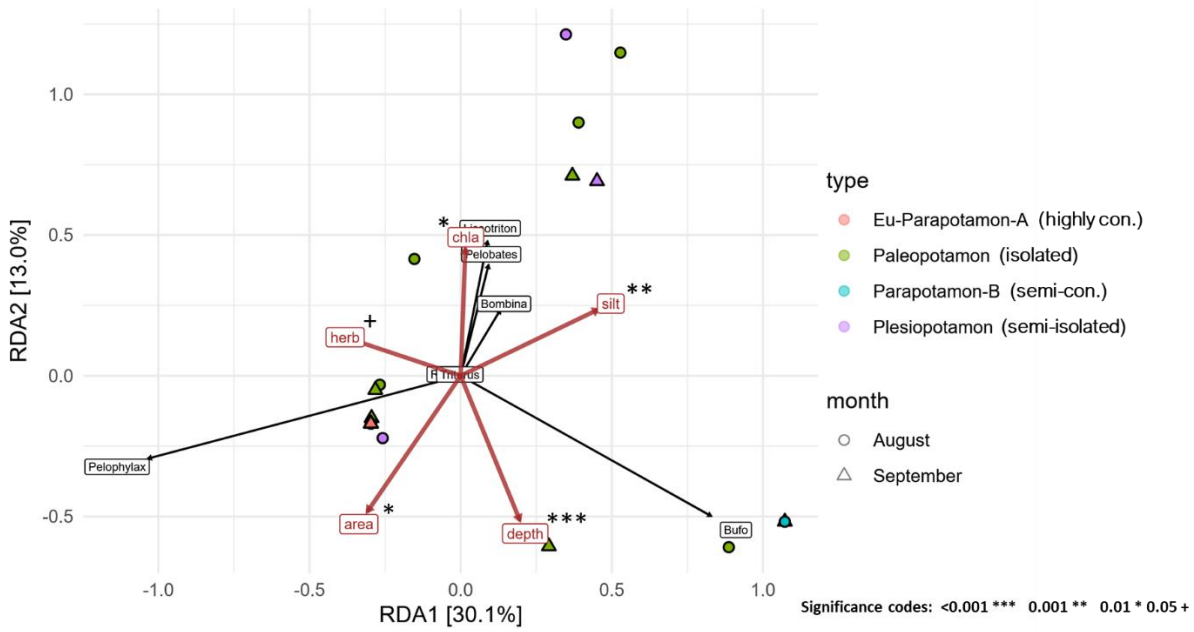

Figure 11: RDA analysis of the most important environmental parameters in correlation with the seven amphibian taxa. The icons represent the sampled sites, while their shape implements the month of sampling (circles in August 2021, triangles in September 2021), and the color shows the type of water body (red = highly connected Eu- or Parapotamon-A, blue = semi-connected Parapotamon-B, lila = semi-isolated Plesiopotamon, and green = isolated Paleopotamon). The plus sign (+) or the number of star signs (*) represents the statistical significance level (see bottom right).

RDA analysis of connectivity parameters (Figure 12) explained 34% of variance on the first axis and 9.9% on the second axis. In more detail, it showed a positive correlation (Table 10) between the three Moran's eigenvector map indices MEM3 (p-value >= 0.003), MEM 5 (p-value = 0.023), as well as MEM 22 (p-value = 0.025) and *B. bufo*, which indicated the scale of community composition patterns depending on the distances between water bodies. MEM10 (p-value = 0.02) showed a positive correlation with *B. bombina*, *L. vulgaris*, and *P. fuscus*. Whereas these three species had a negative correlation with the land-based Euclidean index ‘eu1000 flux’ (with a p-value of = 0.015).

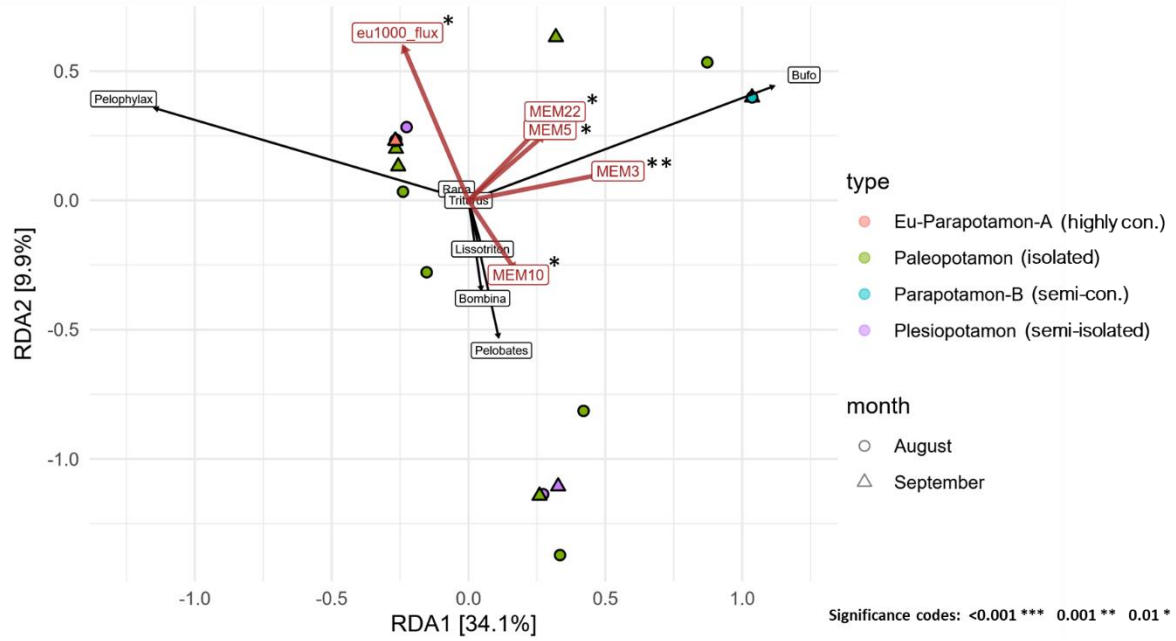

Figure 12: RDA analysis of the most important connectivity parameters in correlation with the seven amphibian taxa. The icons represent the sampled sites, while their shape implements the month of sampling (circles in August 2021, triangles in September 2021), and the color shows the type of water body (red = highly connected Eu- or Parapotamon-A, blue = semi-connected Parapotamon-B, lila = semi-isolated Plesiopotamon, and green = isolated Paleopotamon).

The third RDA analysis of biotic parameters, in particular, fishes (Figure 13) explained 31% of the variance on the first axis and 14.6% on the second axis. In further detail, it showed a positive correlation (Table 11) of the three fish species *Perca fluviatilis* (p-value = 0.004), *Cobitis elongatoides* (p-value = 0.002), and *Scardinius erythrophthalmus* (p-value = 0.033) with *B. bufo*. Further positive correlations were indicated between the genus *Gymnocephalus* and *P. fuscus* (p-value = 0.008), as well as between *Rutilus rutilus* and *B. bombina* (p-value = 0.04). In contrast, *R. rutilus* might have had a slight negative effect on the genus *Pelophylax*. Additionally, Figure 13 shows a clear negative correlation between *Alburnus alburnus* and *L. vulgaris* (p-value = 0.011).

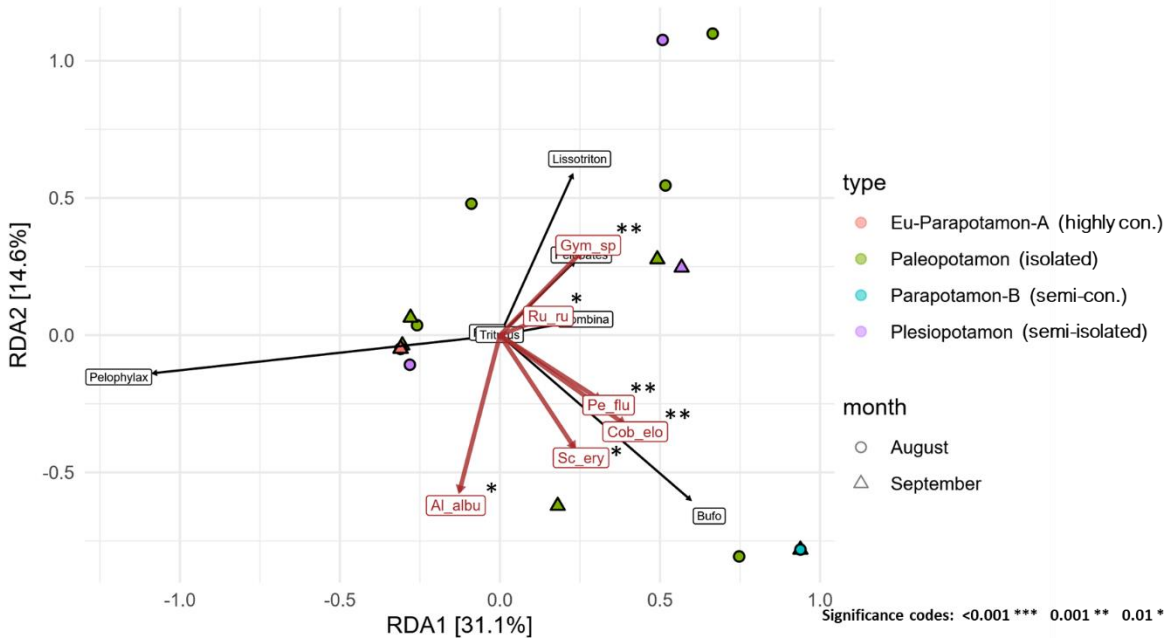

Figure 13: RDA analysis of the most important biotic (fish) parameters in correlation with the seven amphibian taxa. The icons represent the sampled sites, while their shape implements the month of sampling (circles in August 2021, triangles in September 2021), and the color shows the type of water body (red = highly connected Eu- or Parapotamon-A, blue = semi-connected Parapotamon-B, lila = semi-isolated Plesiopotamon, and green = isolated Paleopotamon).

The results of variance partitioning (Figure 14) showed that only 5% of variation could be explained purely by environmental parameters, 10% by biotic (fish) parameters, as well as biotic and environmental parameters combined. The greatest percentage of only a single type of variables (17%) could be explained solely by connectivity parameters. The greatest amount of variance could be explained with the combination of all three types of parameters (environmental, connectivity, and biotic/fishes). The residuals of this model equal 0.39 (39%), which means that the chosen models explained 61% of the variance within the amphibian metacommunity.

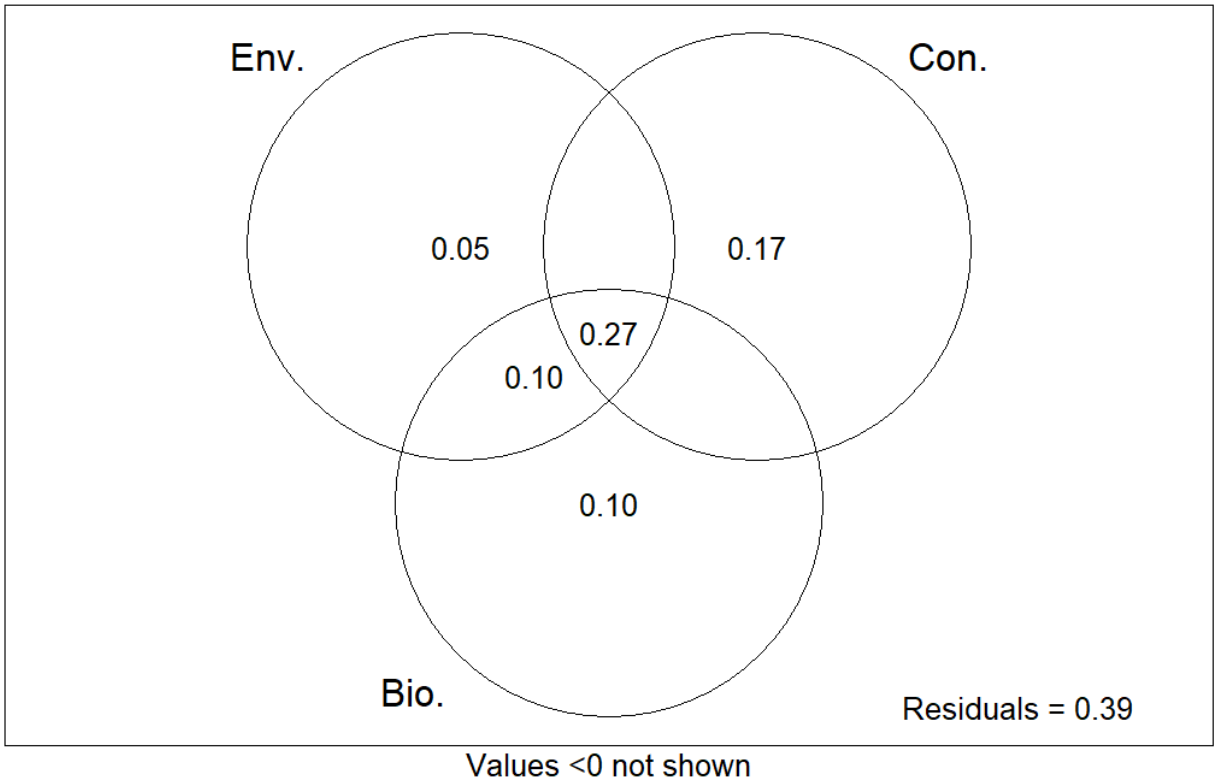

Figure 14: Results of variance partitioning of environmental (Env.), biotic/fish (Bio.), and connectivity (Con.) parameters.

An overview plot of the MEM results is displayed in Figure 15. Braaker et al. (2014) explained that the symbol color reflects the similarity in species composition, whereas the size of the symbols illustrates levels of similarity. In other words, black symbols represent positive eigenvector values and white symbols represent negative values and the size reflects the magnitude (i.e., from -3 to +3) of the values (Hawkins et al., 2007). The different MEMs correspond to communities of different spatial scales, meaning that, for example, MEM1 to MEM7 show large scale community pattern, MEM7 to MEM13 show medium scale patterns, and MEM 14 to MEM19 show small scale amphibian community patterns.

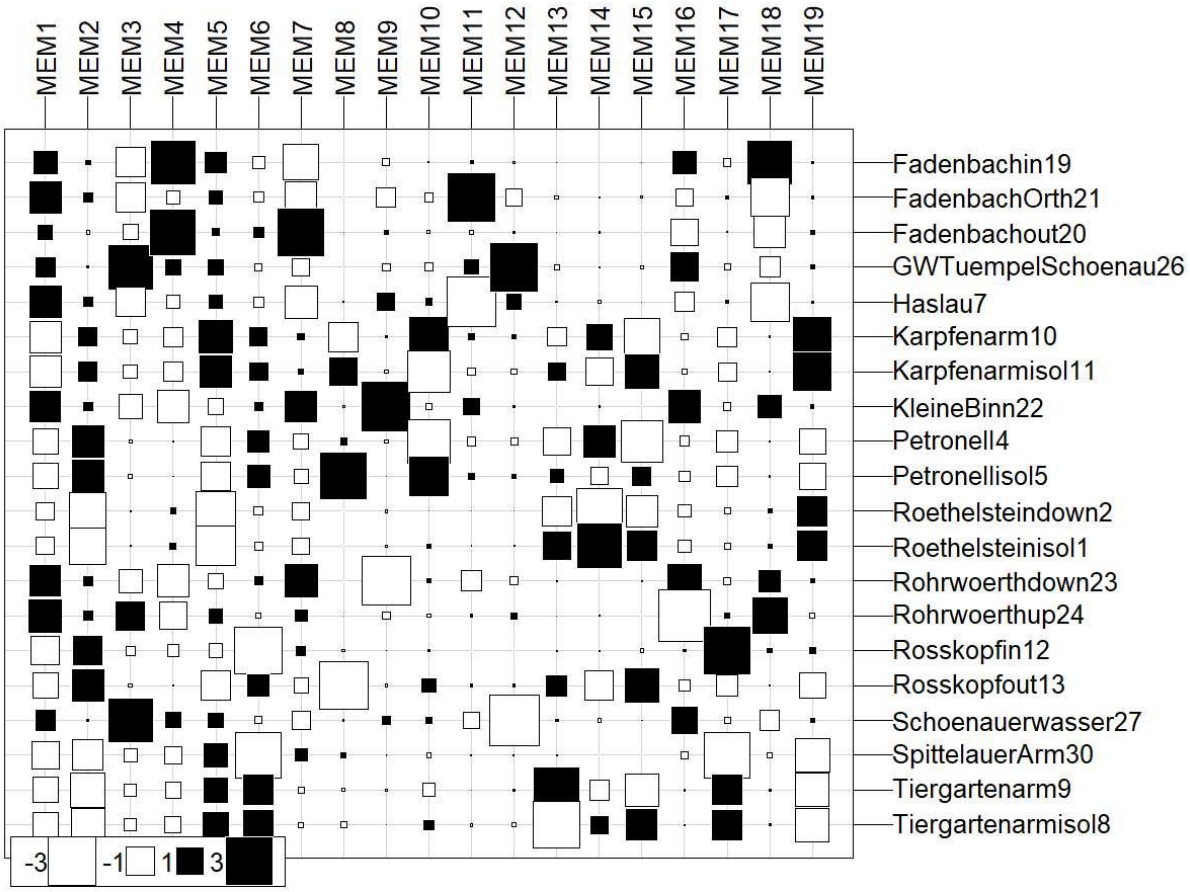

Figure 15: Overview of Moran's eigenvector maps, MEM (Moran, 1948). The color (black or white) of the symbols reflect similarity in species composition, while the size illustrates the level of these similarities (eigenvector values). The different MEMs on the top can be divided into large scaled (e.g. MEM1- MEM7), medium scaled (e.g. MEM7-MEM13), and small scaled (e.g. MEM 14-MEM19) patterns of community composition (Braaker et al., 2014; Hawkins et al., 2007).

Discussion

Amphibian detection via eDNA metabarcoding

The results of this study show eDNA detection of amphibians in 20 out of the 26 sampled water bodies. In more detail, amphibian taxa were detected in 6 of the 8 sampled isolated water bodies and 8 out of 9 sampled semi-isolated water bodies, whereas the greatest number of amphibian taxa (5 and 4) was in two isolated water bodies. On the other hand, with a maximum of one taxon, the lowest number of genera was detected in highly connected (2 out of 5) to semi-connected (in all 4) of the sampled water bodies (including the Danube main channel itself, without detections). Consequently, it can be stated that eDNA metabarcoding is not only suitable for only lentic or lotic waters (Dejean et al., 2012; Lawson Handley et al., 2019; Li et al., 2021; Lopes et al., 2017; Pilliod et al., 2013; Roni et al., 2019; Thomsen et al., 2012; Valentini et al., 2016) but is also valuable to detect amphibians in dynamic and altered floodplain systems. As Lopes et al. (2017) and Valentini et al. (2016) showed, this should be especially true for rare and cryptic species, which is not necessarily the case in this study with only one detection of the specialized and rare species *T. dobrogicus*. However, this might be ascribed to the sampling time which was outside of the spawning season for most amphibians, but in the beginning of their hibernation period (Kogoj, 1997). Furthermore, after Cabela (1990), *B. bombina* was the most abundant species of the Danube floodplains but could only be detected in two water bodies (isolated and semi-isolated) and only in September 2021. This might be due to the fact that they hibernate mainly on land (Kogoj, 1997) and started migration to look for hibernating habitats or dispersed to escape the flood conditions (Tockner et al., 2006) in August 2021. The results in general show differences in detection of occupied water bodies depending on the month (6 of 14 detections merely in one of the months each), which indicate dynamic migration movements likely due to the flood event or the beginning of the hibernation period. It also shows that degradation of amphibian eDNA and thus detection might be restricted to a period of less than 4 weeks in the investigation area during late summer (Lawson Handley et al., 2019; Pilliod et al., 2013; Thalinger et al., 2021). At this point, it is important to emphasize that the goal of this study was not to detect all occurring amphibians in the study area, but rather to test (1) if eDNA metabarcoding is suitable for amphibian detection in dynamic but altered floodplains and to use this method to (2) analyze their distribution and (3) the relative importance of abiotic and biotic environmental conditions, as well as spatial connectivity for amphibian metacommunity composition.

Spatial distribution of amphibians within floodplains

The investigation of this study clearly showed that amphibians occurred along a gradient of aquatic habitats in floodplains according to their traits, as also found by Waringer et al. (2005) among other taxonomic groups (i.e., molluscs, trichoptera, odonatan, and fish). Thus, they created the Floodplain Index (Chovanec et al., 2005; Waringer et al., 2005) to assess the ecological status of 5 described types of floodplain water bodies. This was done by assigning habitat values, as well as indication weights (IW, from 1 to 5; to identify sensitive taxa) to different taxa groups (Waringer et al., 2005). Similar groups of functional habitats have been defined by Funk et al. (2023), as mentioned before (see introduction).

The findings of this study are consistent with several other studies on amphibians, demonstrating that their diversity is highest in mainly isolated to semi-isolated water bodies (Amoros & Bornette, 2002; Larsen et al., 2019; Laufer & Wollenzin, 2011; Marsh & Trenham, 2001; Tockner et al., 1998; Waringer et al., 2005) as mentioned above. Such isolated and semi-isolated water bodies are smaller, less deep, and characterized by more aquatic plants coverage like macrophytes (Chovanec et al., 2005), high chlorophyl-a concentrations in post-connection phases (Weigelhofer et al., 2015), have higher quantities of fine sediment (Funk et al., 2023) and are less likely to contain fish (Amoros & Bornette, 2002). Accordingly, there is less risk of predation and more hiding, spawning, and hibernating areas available (Kačergytė et al., 2021; Kogoj, 1997; Tockner et al., 2006). As a result, these conditions make it more suitable for a broader range of amphibian species like most caudates (i.e., *T. dobrogicus*, *L. vulgaris*) and anurans such as *P. fuscus*, *B. bombina*, and *Pelophylax* species (Kogoj, 1997; Waringer et al., 2005) which could be demonstrated by this investigation. Marsh & Trenham (2001) found effects of pond isolation to be especially important in disturbed environments and theorized that the distribution of terrestrial habitats may be more important to explain such isolation effects than the distribution of breeding habitats (see also Vági et al., 2013). This seems to be congruent with the results in this study, which found only terrestrial connectivity variables to be significantly associated with patterns of species composition or occurrence. Tockner et al. (2006) discovered that rare species – like *T. dobrogicus* in this study – tend to occur only at species-rich sites while common species – like *B. bufo* and *Pelophylax* sp. in this study – tend to be omnipresent in dynamic and more pristine floodplain systems.

Therefore, generalists like the common toad *B. bufo* and water frogs *Pelophylax* sp. also occur in semi- to highly connected water bodies of floodplains. Such connected water bodies are rather large and deep with less aquatic plant coverage (Chovanec et al., 2005; Funk et al., 2023) and are more likely to have fish in them compared to more isolated ones (Amoros & Bornette, 2002). Therefore, only a few amphibians like the common toad *B. bufo* are known to

cope with these environmental conditions (Clausnitzer, 1983; Laufer & Wollenzin, 2011). As a widespread amphibian genus which is strongly linked to water, *Pelophylax* sp. can be expected to occasionally occur in similar water bodies as well (Kogoj, 1997; Nöllert & Nöllert, 1992; Blab, 1986). Both taxa could be detected in large side channel systems which are semi- or highly connected to the Danube River main channel (Chovanec et al., 2005; Funk et al., 2023), although Waringer et al. (2005) would rather expect *Rana* sp. (which were only detected in semi- or isolated sites) to occur there instead of *Pelophylax* species.

Contribution of environmental conditions and spatial connectivity to amphibian metacommunity structure

The outcome of RDA analyses showed a positive correlation of *L. vulgaris*, *B. bombina*, and *P. fuscus* with high content of chlorophyl-a (associated with algal biomass), as well as MEM10 (medium-scaled similarity of amphibian community composition), while they were negatively correlated with surface area and depth of water bodies, as well as with land based Euclidean dPC connectivity of medium distances (i.e., 100-1.000 m). Although chlorophyl-a is mainly correlated with connected water bodies (Amoros & Bornette, 2002; Reckendorfer et al., 2013), this result still seems plausible due to the time of sampling. According to Reckendorfer et al. (2013), chl-a concentrations in isolated water bodies are lowest in the growing season from May until August, while from September to April they are even higher than in connected water bodies. They also found that nutrient concentrations are highest in the main channel water and therefore also high close to its inflow in connected side arm systems and thus, chlorophyl-a from algae also show high concentrations. Hence, the reason why concentration of chlorophyl-a in this study showed a positive correlation with isolated sites associated amphibians (like *L. vulgaris*, *B. bombina*, and *P. fscus*) might be due to inflow of main channel water caused by the flood event (Weigelhofer et al., 2015) in August 2021, which occurred at the end of the growing season of aquatic algae. This is also supported by Amoros & Bornette (2002) who argued that “the effects of hydrological connectivity cannot be reduced to a simple gradient” because of antagonistic effects of connectivity and the complexity of floodplain processes. The negative correlation of some amphibians to large surface areas and depths like the three taxa mentioned above can be explained by their traits (Gabrielsen et al., 2022) and a preference to isolated water bodies (Waringer et al., 2005). *Pelophylax* sp. on the other hand was negatively correlated with silt content, which might be explained by the fact that this genus is bound to a variety of different water bodies from isolated to highly connected water systems with little amounts of silt (Chovanec et al., 2005; Funk et al., 2023; Waringer et al., 2005). In contrast, Waringer et al. (2005) do not expect *Pelophylax* to occur in highly connected sites but rather *Rana* sp. or *B. bufo*. On the contrary to this, *B. bufo* did not occur in Eupotamon and Parapotamon B (highly connected sites) in this study, which might be because this species is

associated with different environmental conditions, spending the summer and winter period mostly in forest areas and migrating longer distances (Kogoj, 1997).

The importance of terrestrial connectivity is also addressed by Jehle (1997) who reported that amphibian dispersal is predominantly terrestrial. This is also reflected by the positive correlation of *B. bufo* with small- and large-scale patterns of MEM indices (e.g., MEM 3, MEM 5, and MEM 22) which are based on terrestrial connectivity with consideration of occupied water bodies as steppingstones (Braaker et al., 2014; Hawkins et al., 2007; Moran, 1948). Consequently, high hydrological connectivity seems to have a negative influence on amphibian metacommunities (Amoros & Bornette, 2002; Chovanec et al., 2005; Tockner et al., 1998, 2000; Waringer et al., 2005). Nevertheless, the dynamic character of river-floodplain systems is essential for amphibian metacommunities to provide a variety of suitable habitats that change over time as explained by the shifting habitat mosaic concept (Stanford et al., 2005). This is supported by the findings of Gabrielsen et al. (2022) stating that spatio-temporal variation in wetlands results in higher amphibian richness and abundance of species and that responses of species to this variability align with predictions based on their natural histories (species traits). Tockner et al. (2006) also stated that the entire floodplain complexity is required to conserve amphibian populations because the exact location of sink and source habitats changes over time with flood events. Fish are known to have a negative impact on amphibians, as Laufer & Wollenzin (2011) reviewed. However, this seems to be contradictory to the findings of this study. In particular, the genus *Gymnocephalus* containing both habitat generalist (Ruffe, *G. cernua*) and specialist species (the Yellow pope, *G. schraetser* and the Danube ruffe, *G. baloni*) (Eros et al., 2024; Waringer et al., 2005) seems to be positively correlated with *P. fuscus,* as well as *Rutilus rutilus* with *B. bombina*, even if these fish taxa are known to have a negative influence on amphibians (Laufer & Wollenzin, 2011). However, the positive correlations between fishes and amphibians in this study could be explained by the simultaneous presence of fishes and amphibians due to an introduction by the previous flood event in summer 2021. Another explanation can be that only eDNA was flushed into semi- to isolated sites which then got detected. Alternatively, it can also be considered that at least some amphibians can withstand the presence of fish for a certain amount of time (Laufer & Wollenzin, 2011). For example, *B. bufo* showed a positive correlation with *Perca fluviatilis, Scardinius erythrophthalmus*, and *Cobitis elongatoides*. This seems plausible, since *B. bufo* is known to be able to cope with the presence of fish due to the production of the toxin bufadienolides present in tadpoles (Hettyey et al., 2019; Kaczmarski et al., 2023). The only clear negative correlation between amphibians and fishes of this study was between *Alburnus alburnus* and *L. vulgaris* which can be supported by several other studies (Hamer et al., 2021; Johnson et al., 2013; Laufer & Wollenzin, 2011) .

Variance partitioning indicated that connectivity (hydrological and terrestrial) alone seems to explain most of the variation (up to 17%), compared to the other variables. This is supported by the findings of Ribeiro et al. (2011) who stated that landscape connectivity alone, without including any biological variables, can provide useful information to predict biological patterns (e.g., amphibian species richness). Fish presence seems to be the second-best explanatory factor of variance (up to 10%), whereas environmental conditions alone explain only up to 5% of the variation. In general, the combination of environmental, connectivity, and biological variables (presence of fishes) is most effective to explain the variation in amphibian metacommunity composition (up to 27%). This means that the amphibian metacommunity is driven by the interaction of local and regional processes that influence the dynamics of environmental conditions and biotic interactions within a landscape as stated by the meta-system theory (Cid et al., 2022). This theory tries to incorporate fluxes of resources, material, and energy into the perspectives of metapopulation and metacommunity ecology in riverine ecosystems that rather focus on spatial flows of organisms. Furthermore, Cid et al. (2022) state that fragmentation can alter local environmental conditions and composition of biota and in addition limit or disrupt fluxes of elements (e.g., water, resources, organisms, energy) with a cascading effect across different organizational levels (like population to ecosystems or even socioecological systems).

Other studies also reflect the complexity of this topic. Tockner et al. (2006) for example found that the metacommunity structure of amphibians differs between active floodplain and riparian forest. Péntek et al. (2016) on the other hand found that mainly local environmental conditions shape the amphibian metacommunity assembly in rainy years, while in dry years spatial signals are more evident, which contradict the results of this study. Gabrielsen et al. (2022) stress the importance for amphibian metacommunity patterns to maintain the historical range of spatio-temporal dynamics within landscapes. Therefore, it is crucial to understand regional processes at each level of organization to effectively conserve, monitor, and restore riverine ecosystems (Cid et al., 2022).

Relevance for conservation, restoration and management

High LCBD values represent the importance for conservation biology of a site and contribution to the overall species composition within a metacommunity (Legendre & Gauthier, 2014). Thus, in this study there are four sites (with a LCBD beta value higher than 0.08) being most relevant for the composition of amphibian metacommunity in the surveyed floodplain system and hence, have high conservation value. Seven other sites seem to indicate a relevant impact on the metacommunity composition as well and therefore show conservation potential (with LCBD beta values between 0.04 and 0.08). On the contrary, the other 17 sites have

considerable potential for ecological restoration. Furthermore, Tockner et al. (2006) state that levels of nestedness should be considered as an indicator for conservation because rare species tend to occur only at species-rich sites. Similarly, Nagel et al. (2021) recommend prioritizing high-productivity pools, as well as clusters of water bodies along a range of hydrological conditions for amphibian conservation (Gabrielsen et al., 2022), especially in the face of increasingly variable climate. Additionally, degrees of urbanization should be considered within a 1km buffer of water bodies, since it is universally associated with substantial changes in wetland characteristics and a negative effect on aquatic diversity and species assemblage (Johnson et al., 2013).

Integrating eDNA metabarcoding together with spatial modeling approaches (e.g., Funk et al., 2023; Gabrielsen et al., 2022) thus provides an effective tool for broad-scale biodiversity assessment in conservation and restoration management. Therefore, on a general basis, the implementation of the presence-absence based floodplain index (Chovanec et al., 2005; Waringer et al., 2005) according to the requirements of the EU WFD (Water Framework Directive) by using cost-effective eDNA metabarcoding is recommended, especially because there is still no standardized method to assess the ecological status of the Danube River and its floodplains (see Ofenböck et al., 2016). Additionally, this would help with gaining a deeper knowledge of amphibian metacommunities, as there are still only a few long-term studies on them (Jehle, 1997). Including amphibians in the ecological assessments is especially important to holistically understand habitat requirements necessary for all biota and to counterbalance conservation implementations in favor of more dynamic situations (e.g., when only based on fishes) (Chovanec et al., 2005). As mentioned above, the floodplain index is based on the assignment of habitat values and indication weights to different taxa groups occurring in 5 water body types (Chovanec et al., 2005; Waringer et al., 2005).

These types of water bodies, like in this study (also see Eros et al., 2024; Funk et al., 2023) require different measures of conservation and restoration. Funk et al. (2023) for example, stated that highly connected sites are most relevant to reconnect the whole system to the main channel which causes higher erosion and deposition rates after the reconnection, while semi-connected sites are important for relocation of substrate and rejuvenation after the reconnection and therefore represent the most dynamic zone in a floodplain. This is congruent to the findings of Amoros & Bornette (2002) that river dynamics can compensate for the effects of succession which leads to a decline in surficial hydrological connectivity creating more heterogeneity in floodplain. This restart of ecological succession at a regional scale is important (Amoros & Bornette, 2002), as semi-isolated sites are most threatened by sedimentation and have a short life span due to strong aggradation processes (Funk et al., 2023). Here, active sediment management (e.g., by excavation), sediment trapping for inflowing water, as well as

flood water management (e.g., by weirs) can be effective measures, as an upstream reconnection is complicated due to their position in the floodplain system and hence, would be more problematic (Funk et al., 2023). Isolated sites, on the other hand, are less threatened by sedimentation processes as they already took place, making these water bodies more persistent but also more dependent on groundwater levels (Funk et al., 2023). This, and many other studies showed, that these are the most important water bodies for amphibians (Amoros & Bornette, 2002; Kogoj, 1997; Larsen et al., 2019; Laufer & Wollenzin, 2011; Tockner et al., 1998; Waringer et al., 2005), making it especially important to protect them from internal terrestrialization processes, as well as from the lowering of groundwater driven by climate change, river bed incision, and human alterations (Alford & Richards, 1999; Cardinale et al., 2012; Funk et al., 2023; Marsh & Trenham, 2001; Nori et al., 2015; Schindler et al., 2016; Tickner et al., 2020). To achieve this, measures that raise groundwater levels like artificial water supply or removal of sediment (e.g., via excavation) would benefit such isolated water bodies on a local scale (Funk et al., 2023). On a regional scale, it would be beneficial for a broad spectrum of biota to restore historical spatio-temporal dynamics in floodplain systems to conserve these extraordinary diverse ecosystems (Amoros & Bornette, 2002; Chovanec et al., 2005; Cid et al., 2022; Gabrielsen et al., 2022; Nagel et al., 2021; Tockner et al., 2006).

Conclusion

Although there are clear limitations to the obtained data due to a low eDNA sequencing depth (total number of sequence-reads per site), valuable insights could be gained about amphibian metacommunity composition in (fragmented) floodplain systems by using eDNA metabarcoding. The results lead to the conclusion that the amphibian metacommunity is driven by the interaction of local (abiotic and biotic environment) and regional (connectivity) processes (Cid et al., 2022), while terrestrial connectivity seems to be the most dominant driver (Marsh & Trenham, 2001; Vági et al., 2013). Furthermore, the findings show that most amphibian taxa in floodplain systems like the one of the Danube River (especially *B. bombina*, *P. fuscus*, and *L. vulgaris*) occur in hydrologically disconnected, small, shallow, stagnant water bodies rich in aquatic plants but mostly without fish and with low levels of hydrological connectivity (Amoros & Bornette, 2002; Chovanec et al., 2005; Funk et al., 2023; Kogoj, 1997; Larsen et al., 2019; Laufer & Wollenzin, 2011; Marsh & Trenham, 2001; Tockner et al., 1998; Vági et al., 2013; Waringer et al., 2005). Yet, there are some exceptions like *B. bufo* that withstands the presence of fish (Hettyey et al., 2019; Kaczmarski et al., 2023; Laufer & Wollenzin, 2011) and occurs in open-water bodies that are easy to reach via terrestrial migration (Kogoj, 1997). Another example is the genus *Pelophylax* which is strongly bound to water (Kogoj, 1997), but occurs in a variety of permanent water bodies (Waringer et al., 2005) with structural features like woody debris or aquatic plants (Vági et al., 2013), whereas the presence of fish is not that relevant. Including amphibians in ecological status assessment, for example, by implementing the floodplain index in the European WFD by using eDNA metabarcoding, would counteract conservation implementations in favor of more hydrologically connected conditions (Chovanec et al., 2005). Furthermore, it would contribute more data on amphibian metacommunities, as there are only a few long-term studies on that matter (Jehle, 1997). The gained knowledge could also contribute to a better understanding of the functioning of dynamic but fragmented ecosystems like the Danube floodplains (Cardinale et al., 2012; Funk et al., 2019; Schindler et al., 2016; Tickner et al., 2020). This would lead to improvements in conservation, restoration, and decision making in management, which are urgently needed in these highly altered and threatened ecosystems worldwide. This study showed that the non-invasive and cost-effective eDNA metabarcoding method can be a helpful and comprehensive tool to achieve this, and can further decipher the fascinating complexity of amphibian metacommunity ecology.

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